SPATIAL AND TEMPORAL ANALYSIS OF LAND COVER, CLIMATE, AND LAKE WATER QUALITY IN THE MATANUSKA-SUSITNA, VALLEY, ALASKA

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Abstract

The Matanuska-Susitna (Mat-Su) Valley in south central Alaska is a rapidly developing and warming region with a high density of lakes. Water quality in these lakes is a growing concern because of rapid land use change in a valley influenced by wind, glaciation, and complex groundwater-lake hydrology. To monitor changes in lake water quality, the Matanuska-Susitna Borough implemented a long-term lake monitoring program with citizen-scientists conducting water quality sampling monthly since 1998 in over 50 lakes. Results of this monitoring showed that total phosphorus decreased, specific conductivity increased, and other parameters remained unchanged in lakes across the Mat-Su Valley (n=17). Lakes with steeper watersheds and higher proportions of Knik Silt Loam had higher specific conductivity. This suggests that aeolian deposited silt high in clay content from the Matanuska braid plain interacts with ground and surface water to contribute to high specific conductivity in lakes. Lakes experiencing higher rates of forest loss due to residential development in their watershed increased more in specific conductivity, suggesting runoff from impervious surfaces increased the dissolved ion load in lakes. Increases in specific conductivity also coincided with increases in the amount of total wind, suggesting wind deposited silt and wind driven waves both suspended and increased the number of dissolved ions in the water column. The more frequent occurrence among lakes of an increasing trend in specific conductivity during August, September, and October supports this claim, as aeolian transportation of silt from the Matanuska braid plain is most likely at its peak during those times. This study provides a baseline sample of water quality parameters for making decisions in the lake rich landscape of the Mat-Su Valley.

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1 Introduction

There are more than 5,000 lakes in the 65,000 km^2 Matanuska-Susitna (Mat-Su) Borough (MSB) and more than 400 in the Core Planning Area of the Mat-Su Valley, where most people live. The Mat-Su Valley lake district makes up one of 20 lake districts in Alaska based on high lake abundance (Arp 2009). It is unique among lake districts in Alaska due to its relatively large population. The most recent population growth and corresponding development in the Mat-Su Valley occurred in the late 1990's to mid-2000's when the MSB began a long-term, citizen-scientist based monitoring program that has now spanned 16 years and monitored more than 50 lakes. Concern for impaired lake water quality arose because of rapid development, which sparked studies at Big Lake (Woods 1992), Lake Lucille (ADEC 2001) and several studies of individual lakes in the Mat-Su Valley. Despite continued growth, no previous study examined geographic or temporal trends among lakes and their water quality in the Mat-Su Valley lake district, which is the goal of this thesis.

Land cover is an important indicator of in-lake processes (Shannon 1972, Baker 1985 & Gove 2001). Land cover change is a metric often used to quantify development in a geographic context. This allows for hypothesis testing at large geographic scales with respect to the effects of anthropogenic disturbance on water quality. Increased urban land cover can alter hydrologic responses to precipitation events in watersheds by changing the timing and quantity of surface runoff (Mateus 2015, Sajikumar 2014), potentially impairing surface water quality (Mattikalli 1996, Choi 2003, Wilson 2010). Examples of impairment include increases in nitrogen, phosphorus (Tong 2002), specific conductivity, and acidity of water (Conway 2007).

Often the first indication of pollution in lakes is an increase in specific conductivity, the basis for measures of total dissolved solids and salinity. Specific conductivity (SC) is the measure of how fast electrical current can travel through water at 25° C and depends on the total cations and temperature of the water. Spatial variability of specific conductivity has been shown to depend on the composition of soils and surficial geology in the watershed (Miller 1988). In Green Bay, Wisconsin, higher SC was strongly correlated with loading from the Fox River and near-shore urban land cover classes (Yurista 2015). An example from Minnesota, another lake rich region, found that water clarity was lower in urban and agricultural regions than in predominantly forested landscapes (Olmanson 2014). The changes occurring in lakes in the 48 contiguous states are suggestive of future potential change in the Mat-Su Valley as populations increase. This suggests that water quality monitoring in lakes be made a priority so that managers and planners can avoid negative impacts to lake water quality and to establish baseline conditions now, in the early stages of rapid growth in the Mat-Su.

In addition to expanding populations, the Mat-Su has experienced a significant increase in temperatures over the past century, an increase projected to continue (McAfee 2014). Changes associated with a warming climate may lead to changes in the timing and intensity of wind, rain, and snowfall, altering hydrologic and physical processes that determine lake water quality (Prucha 2011). Additional effects of changes in the hydrologic cycle may lead to in-lake changes in ice phenology, water balance, energy balance, evaporation, inflow, algae bloom timing and duration, and zooplankton community composition (Duguay 2003, Brown 2003, McCabe 2007, Prowse 2011, Carter 2012, Arp 2013). Changes in SC can occur from imbalanced evaporation to inflow ratios driven by warm and dry conditions, as documented in the Yukon Flats of interior Alaska, where salinity has significantly increased in lakes that are shrinking (Lewis 2014).

In the years since the implementation of Mat-Su Borough's citizen scientist driven program in 1998, the population has doubled (US Census Bureau, 2010 Census). The program has monitored more than 50 lakes, some for as many as 16 seasons. Long-term monitoring of surface water quality is time and resource intensive. Implementing a citizen-based program is one of the most successful ways to implement long-term scientific research and conservation of natural resources (Lottig 2014, Link 2006, Wilson 2013, Burt 2008).

The purpose of this thesis is to take a broad geographic approach to evaluating temporal and spatial trends in water quality among lakes in the Mat-Su Valley. Factors considered here include potential negative impacts to water quality due to rapid development and climate change as well as the scarcity of long-term lake studies. The primary goals were twofold: to establish a baseline for the lakes that have been monitored and identify factors, including development, that likely determine lake water quality. These goals of a baseline and identification of important factors will assist planners and managers in maintaining water quality in the face of future development and climate change. The research conducted for this thesis evaluated (1) the spatial variability of water quality among lakes in the Mat-Su Valley, (2) temporal trends in water quality for a subset of lakes, (3) the key landscape, hydrologic, and in-lake morphometric variables related to water quality, and (4) landscape and climatic changes affecting water quality trends.

2 Materials and Methods

This thesis reports on the evaluation of lake water quality, as well as the trends and possible effects of environmental variables on lake water quality in five sections:

(1) Water quality parameters summarized for 46 lakes.

(2) Trend analyses of median water quality parameters from lakes with more than 5 years of data (n=17) to identify annual and monthly trends.

(3) Geospatial analysis of landscapes surrounding lakes (n=41) to evaluate relationships among water quality, the landscape, hydrology, climate, and in-lake morphometry.

(4) Multivariate statistical methods identifying the most important landscape variables determining water quality.

(5) Analyses of effects on trends in specific conductivity (SC) from a subset of 15 geographically independent lakes with more than 5 years of water quality monitoring data.

2.1 Study Location: Northern Cook Inlet

The lakes in this study are located in northern Cook Inlet Ecoregion in Southcentral Alaska and are within the Susitna or Matanuska basins (Figure 1). The ecoregion extends west to the Alaska Range, east to the Matanuska Valley between the Talkeetna Mountains and Chugach Mountains, north of Talkeetna and south to the Kenai Peninsula. The Cook Inlet Ecoregion is characterized by rolling terrain ranging in elevation from sea level to 600 meters above sea level (Gallant 1995).

The mild but transitional maritime and continental climate of this ecoregion has attracted a majority of human development (Gallant 1995, Bieniek 2012). Average temperatures range from -10° C in the winter to 21° C in the summer, and precipitation ranges between 39 cm per year in Palmer and 71cm in Talkeetna. Ice and snow cover typically last from October to April and strongly influence hydrologic processes (Kikuchi 2013).

Remnants of Pleistocene glaciation, such as moraines, drumlin fields, eskers, outwash plains, and very large dunes (VLD), characterize the surficial geology of the region (Weidmer, 2011). A variety of vegetation types occur in the Cook Inlet Ecoregion, including widespread needleleaf, broadleaf, and mixed forests dominated by white spruce (*Picea glauca*), paper birch (*Betula papyrifera*), and balsam poplar (*Populus balsamifera*). In the floodplains, tall scrub form thickets dominated by alder (*A. tenuifolia, A. crispa, A. sinuata*). Low scrub communities occupy the poorly drained lowlands with tall scrub swamp, low scrub bog, wet forb herbaceous, and wet graminoid herbaceous in the wettest sites (Gallant 1995).

Human settlement of the Mat-Su Valley began hundreds of years ago but has increased substantially only over the last half-century. This region of the Mat-Su Valley was originally inhabited by the Dena'ina and Ahtna Athabaskans, who called the area around Wasilla "Benteh" which translates as "among the lakes" (Bright 2004). Subsequently, Russia settled the valley in the 1700's, followed by United States settlement in the early 20th century. The relatively recent settling of lands in the Mat-Su Valley brought visible changes to the landscape through farming, development, and mining. Much of the 101,000 people who reside in the borough currently live in the Core Area of the Mat-Su Valley within and between the towns of Palmer, Wasilla, and Houston (Kikuchi 2013), giving the Mat-Su Valley a distinctly rural character.

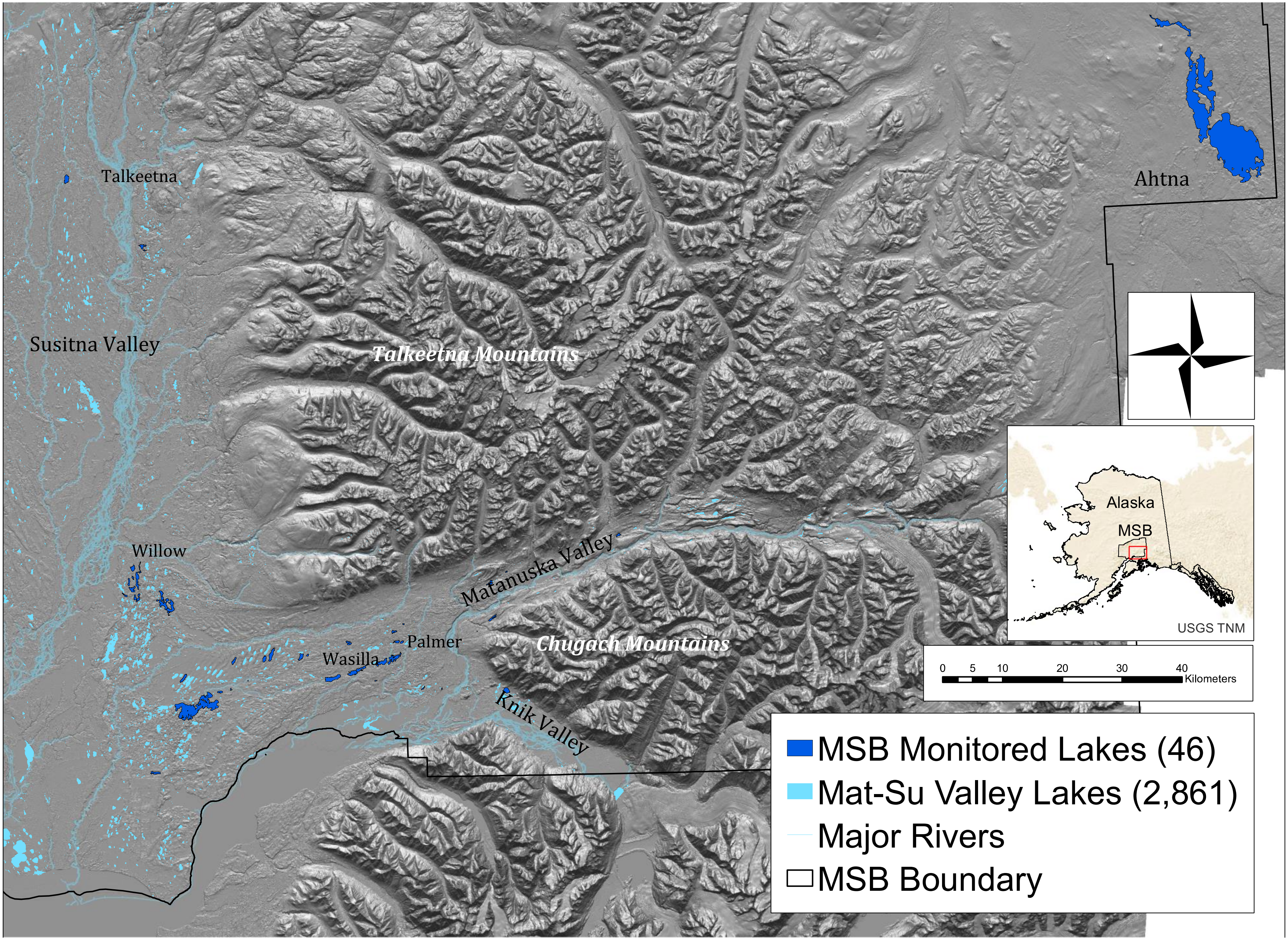

Figure 1. Lakes (n = 46) monitored for water quality from 1998-2014 in the Matanuska-Susitna Borough (MSB)

Between the 2000 Census and the 2010 Census population density grew from 2 to 3.6 people per square mile. In comparison, population densities for the towns of Palmer and Wasilla are 1,206 and 467 people per square mile, respectively (U.S. Census 2010). Other smaller communities in the Mat-Su Valley include Big Lake, Willow, Houston, Knik-Fairview, Talkeetna, Chickaloon, and Butte. Approximately 98% of land use in the Mat-Su Borough is vacant land, followed by residential development at 1%.

Residential development in the Mat-Su Valley occurred around the lakes first and spread outward as lakes have provided significantly higher property values (Berman 2013). Jokela et al (1990), identified and classified 439 lakes in the Mat-Su Core Area, where most of the lakes in this study are located. In the Core Area 351 lakes (80%) are seepage lakes with no inlet or outlet, 56 (13%) are drainage lakes with inlets and outlets, and the remaining 32 (7%) have either inlets or outlets only. Most lakes are low in primary productivity, but differ in specific conductivity, alkalinity, pH, and total nitrogen depending on hydrologic type (Jones 2003).

2.2 Matanuska-Susitna Valley Lake Sampling

Citizen-scientists measured seven common water quality parameters every month of the ice-free season in over 50 lakes, some for as many as 16 years, as part of the Matanuska-Susitna Volunteer Lake Monitoring Program (MSB-VLMP). Specific conductivity (SC; $\mu S\ cm^{-1}$), was measured with a Quanta Hydrolab and corrected to 25° C. It was measured on a profile at every meter of depth at the same location each time as marked by GPS along with pH, temperature (T; °C), and dissolved oxygen (DO; $mg\ l^{-1}$). Measures of lake productivity were taken at the time of profile measurements of total phosphorus (TP;

μg l-1), chlorophyll-a (CHL-a; μg l^{-1}), and sechhi-depth (SD; m). TP and CHL-a samples were collected independently from a depth of 2 meters using a Van-Dorn Sampler. Both samples were marked with the date, time, location, name of the sampler, and put on ice to be delivered to the lab where they were filtered. TP samples were put in a 1 liter glass bottle and EPA method 365.3 is used to determine TP (EPA 1978). CHL-a samples were fixed with Hydrochloric acid in a 1 liter plastic bottle in the field to be filtered at the lab and concentrations determined using Standard Method 10200 H (APHA 1998). Two secchi-depth measurements were taken on the shaded side of the boat if applicable, once on the descent of the secchi-disk and once on the ascent, then averaged. Volunteers organized by the MSB-VLMP have conducted the lake monitoring since 1998. The Matanuska-Susitna Borough Watershed Coordinator and staff have certified returning and new citizen-scientists in sampling techniques each year and calibrated the Quanta Hydrolabs annually. Data is input into excel spreadsheets and checked by Mat-Su Borough staff for quality control.

2.3 Water Quality Data Management and Summary

The water quality data was provided for this study to evaluate the quality of the data and initiate the analysis of spatial and temporal trends. Lake water quality data was evaluated for anomalies, cleaned by removing erroneous outliers, and non-detects were assigned a value of half the detection limit of the test being used. After confirming the high standard of data, the mean, median, minimum, maximum, and standard deviation were calculated for each lake for all water quality parameters (Table 8, Appendix).

2.4 Water Quality Trend Analysis

Using a subset of 17 lakes that had at least 5 years of water quality data with at least three sampling sessions per ice-free season (May-October), average annual trends and monthly trends were evaluated. Annual trends evaluated multi-year trends in values averaged across the entire ice-free season. Monthly trends evaluated multi-year trends averaged across a given month. Median values were used to reduce the influence of outliers. Of the lakes in this study (n=17; Table 1), eight lakes were seepage lakes with no outlets or inlets and nine were discharge lakes with at least one outlet or inlet. Lakes were classified in this way according to their dominant hydrologic input based on the NHD and following the method of Jones et al (2003). Each water quality variable, chlorophyll-*a* (μg L^{-1}), total phosphorus (μg L^{-1}), specific conductivity (μS cm^{-1}), temperature (°C), dissolved oxygen (mg L^{-1}), and pH, were tested for each lake for the presence of an upward or downward trend over time with linear regression using R Statistical Software. To evaluate average trends across the Mat-Su Valley, average annual trends were calculated for each lake and each water quality parameter from the slope of the linear regression. Then the standard error was calculated and multiplied by 1.96 (95% Confidence Interval), first added to the mean then subtracted from the mean. If both numbers were negative, it indicated an average decrease in a water quality parameter across all lakes in the Mat-Su Valley. If both numbers were positive, it indicated an average increase in a water quality parameter across all lakes in the Mat-Su Valley. To evaluate upward or downward trends happening during a particular month, linear regression was used for each water quality variable for each month of the ice-free season from May to October over the years for each lake.

The lakes analyzed for temporal trends ranged in size from 1010 ha (Big Lake) to 1.6 ha (Juke's) with the mean surface area (SA) being 103 ha. Mean depths ($\bar{z}$) ranged from 11 m (Big Lake) to 0.36 m (Wolverine) with a mean of 5.5 m. Six lakes did not have bathymetric data available therefore volumes and mean depths were not available. The known volumes (V) range from 1.12 x 10^7 m^3 (Big Lake) to 1.47 x 10^5 m^3 (Memory), the mean being 1.04 x 10^7 m^3. Water quality data extended back to 1998 on some lakes with up to 67 monitoring sessions over 15 years. Juke's lake has the fewest monitoring sessions with 14 over 5 years. The mean number of monitoring sessions is 31 and an average temporal scale of 10 years.

Table 1. Morphological characteristics of 17 study lakes studied for temporal trends.

Lake	Hydrologic regime	Mean Depth (m)	Max Depth (m)	Perimeter (km)	Surface Area (ha)	D_L
Big Lake	Discharge	11.08	27.1	61.64	1009.7	4.67
Birch	Seepage	NA	3.9	1.26	5.98	1.45
Carpenter	Seepage	2.47	9.1	5.13	71.2	2.04
Dandy	Seepage	NA	10.8	0.89	4.89	1.13
Finger	Seepage	4.73	13.4	13.57	146.5	3.21
Fish	Discharge	NA	16.5	2.45	54.33	1.44
Florence	Seepage	5.32	12.5	2.93	22.3	1.54
Hulbert Pond	Seepage	NA	2.9	0.58	1.63	1.29
Juke's	Discharge	NA	2.9	0.94	2.32	1.74
Little Question	Discharge	NA	15.9	0.98	6.21	1.11
Lucille	Discharge	1.73	6.1	6.88	146.5	1.57
Memory	Seepage	2.2	6.1	4.04	34.0	1.93
Morvro	Seepage	3.31	5.2	3.23	35.2	1.47
Nancy	Discharge	7.66	19.8	26.97	308.0	4.24
Neklason	Discharge	4.79	17.4	3.04	29.1	1.62
Seymour	Discharge	2.14	5.8	5.85	92.7	1.69
Wolverine	Discharge	0.37	2.1	5.46	40.1	2.43

2.5 Landscapes around lakes

To determine which landscape factors affecting lake water quality, analysis of spatial variability in water quality among 46 lakes at two scales: the watersheds and local scale (within 100 meters of lake shorelines). These lakes were selected because each had at least two years of monitoring with at least three sampling sessions conducted during the ice free season for each year. Water quality was summarized by median values for each lake and put in a GIS for spatial variability analysis. The landscape variables were calculated in ArcGIS 10.2.2. and linked to the lakes in the GIS. The first set of landscape variables were calculated for the area within a 100-meter buffer of the lake shoreline. The second set of landscape variables were calculated at the scale of the contributing watershed, delineated with Terrain Works NetMap add-in in ArcGIS 10.2.2. with 5 x 5 meter Digital Elevation Model (DEM).

Landscape variables consisted of various anthropogenic and geomorphic variables. Land cover data and land cover change data were available for Alaska from the U.S. Geological Survey (USGS) Multi-Resolution Land Characteristics Consortium (MRLCC) National Land Cover Database 2011 (NLCD2011) and NLCD 2001 to 2011 Land Cover Change (NLCDC), each at 30 x 30 meter pixel resolution (www.mrlc.gov/nlcd11_data.php). NLCD classification is modified from the Anderson Land Cover Classification System with 20 classifications (Anderson 1976, Homer 2015). Mean hillslopes (%), elevation (m), and total area (ha) of the watershed and 100-m buffer were calculated using a 5x5 m DEM. Lake morphological data was obtained from the Mat-Su LiDAR and Imagery project database (http://www.matsugov.us/it/gis/2011-lidar-imagery-project) and lakes were classified as discharge or seepage according to stream connectivity based on the USGS's National Hydrography Database (NHD).

Temperature and precipitation data for lakes were determined by the closest weather station (Menne 2012, NOAA, accessed May, 2015). Mean annual temperature and total annual precipitation from seven weather stations were averaged for the 2001-2011 time period and the 1981-2000 time period, then differenced and determined for lakes and put in the GIS. Temperature and precipitation data were available from the National Climatic Data Center (NCDC).

2.6 Lake Water Quality – Landscape Relationships

A global multidimensional scaling using the function monoMDS in R statistical software was applied to median lake water quality parameters of 46 lakes. Median water quality parameters were transformed by subtracting the mean from each value, dividing by the standard deviation, and adding 10 to ensure there were no negative values since monoMDS requires positive values. A Manhattan distance measure was used and dimensionality was assessed by creating a scree plot of the number of stress on the number of dimensions. The number of dimensions was selected by determining the largest reduction in stress between dimensions, or the number of dimensions where the plot forms the elbow (McCune 2002). Three dimensions were used in the final solution. Two randomized runs were used to produce the solution. Stability of the solution was assessed by selecting a solution where the residual mean standard error and max residuals were less than 10^{-4}.

Additionally, a principal components analysis (PCA) was conducted using all landscape and water quality variables to determine landscape variables that covary with median water quality. Median water quality parameters and landscape variables were transformed by subtracting the mean from each value, dividing by the standard deviation.

Spearman Rank correlation used to verify correlations between all lake water quality variables and all landscape variables calculated in the GIS. The non-parametric test was used because it can account for non-linear relationships.

2.7 Land cover change, climate change and specific conductivity

To determine the best predictors of specific conductivity change in lakes, land cover change from 2001-2011 were calculated for the 100-m buffer and watershed. Temperature and precipitation differences between 1981-2000 and 2001-2011 were calculated from the seven NOAA weather stations in close proximity to the lakes. Wind speed and direction were not analyzed due to scarce wind measurements across the Mat-Su Valley. Average specific conductivity change was calculated for the 15 lakes by calculating the slope of the linear regression trend from median annual water quality parameter and multiplying that slope by the number of years data were collected in each lake. Land Cover change, temperature change, precipitation change, lake morphological values, and mean hillslope were used in a backward stepwise multiple regression evaluated by Aikike Information Criterion (AIC) to test the effects on specific conductivity change.

3 Results

3.1 Summary of Lake Water Quality

Lakes monitored in the Mat-Su Borough (Figure 1) by citizen-scientists during the ice-free season from 1998-2014 were generally of good water quality (Table 1). Temperature, dissolved oxygen (DO), specific conductivity (SC), pH, chlorophyll-a (CHL-a), and total phosphorus (TP) were below thresholds set by the State of Alaska Department of Environmental Conservation and the EPA (ADEC 2012, EPA 2013). Average water temperatures were low, with maximum temperatures less than 25°C and as low as 0°C from May to October. Average DO was generally suitable for fish habitat but minimum levels reached as low as 0.04 mg L-1 and maximum concentration exceeding 22 mg L-1. Specific conductivity was generally lower than 200 µS cm-1, with no dissolved ions detected in some lakes and the highest SC measurement over 5000 µS cm-1, about half the average levels measured in wastewater sewage. Lakes were neutral to alkaline. The lowest pH was 5.27 and highest was 10.20. According to average CHL-a, lakes were oligotrophic with multiple lakes below detectable levels. The highest CHL-a concentration was 22 µg L-1. Average TP concentrations indicate mesotrophic lakes with the lowest TP below detectable levels, and the highest TP at 283 µg L-1.

Table 2. Summary statistics for ice-free seasons (2001-2014) of temperature, dissolved oxygen (DO), specific conductivity (SC), pH, chlorophyll-a (CHL-a), total phosphorus (TP) and morphometry for all 46 study lakes.

Lake Variable	Mean	Median	SD	Minimum	Maximum
TP (µg L^{-1})	11.37	11.25	5.39	0.00	22.50
CHL-a (µg L^{-1})	1.96	1.60	1.20	0.40	6.00
SC (µS cm^{-1})	151.75	136.00	145.87	8.00	769.00
pH (SU)	7.84	7.85	0.42	6.93	8.65
DO (mg L-$^{-1}$)	9.34	9.42	1.91	1.11	14.49
Temperature (°C)	13.58	14.46	3.71	5.57	19.79
Perimeter (km)	8.61	3.23	15.95	0.29	80.20
Maximum Depth (m)	10.01	6.55	9.42	1.50	47.20
Surface Area (ha)	290.71	37.44	1027.48	0.48	5990.55

3.2 Water Quality Trends

3.2.1 Annual trends

On a lake-by-lake basis median annual (averaged over all values from a given year) water quality from 2001 to 2014 was generally unchanged in most lakes with the exception of SC and TP (Figure 11, Table 3). Specific conductivity generally increased in most lakes. Four of the 17 lakes showed statistically significant linear increases in median SC, 7 lakes increased overall, 7 remained unchanged, and three decreased. Specific conductivity increased in 75% of lakes (Mean trend – (Stantdard error*1.5)). Fish, Nancy, and Little Question lakes decreased enough to adjust the mean trend minus the standard error to be lower than zero, indicating no significant trend at the 95% confidence interval but significant at the 75% confidence interval. Seepage lakes generally increased (three significantly so) while one discharge lake decreased and another increased significantly in median SC. The largest total increases in SC were in Lucille (n = 8, 101 μS cm^{-1}), Hulbert (n = 5, 89.40 μS cm^{-1}), Finger Lake (n = 14, 60.9 μS cm^{-1}), Seymour (n = 15, 50.7 μS cm^{-1}), and Big Lake (n = 11, 42.7 μS cm^{-1}). The significant increases in SC in Memory, Seymour, Lucille, and Big Lake coincided with moderate decreases in pH over the same time period.

TP decreased on average in lakes across the Mat-Su Valley (-0.72 mg yr^{-1}), though none changed in trophic state, . In fact, only on seepage lake and two discharge lakes actually showed increasing TP trends. In contrast, the number of lakes showing increasing trends nearly equaled the number of lakes with decreasing trends in CHL-a, temperature, DO, and pH.

Table 3. Average annual trends calculated from the mean of all lake trends from 6 water quality parameters for lakes (n=17) in the Mat-Su Valley from 2001-2014.

	CHL-a (μg l^{-1})	TP (μg l^{-1})	Temp (°C)	DO (mg l^{-1})	pH	SC (μS cm^{-1})
Mean Trend	0.01	-0.72	-0.02	-0.03	0.02	1.85
Standard Error	0.06	0.23	0.06	0.06	0.01	1.16
Upper 95%	0.13	**-0.28**	0.10	0.08	0.04	**4.11***

Lower 95%	-0.10	**-1.17**	-0.15	-0.15	-0.01	**-0.42***
Average Trend	no trend	decreased	no trend	no trend	no trend	increased*

***Specific conductivity increased in 75% of lakes (Lower 75%: Mean-SE*1.5).**

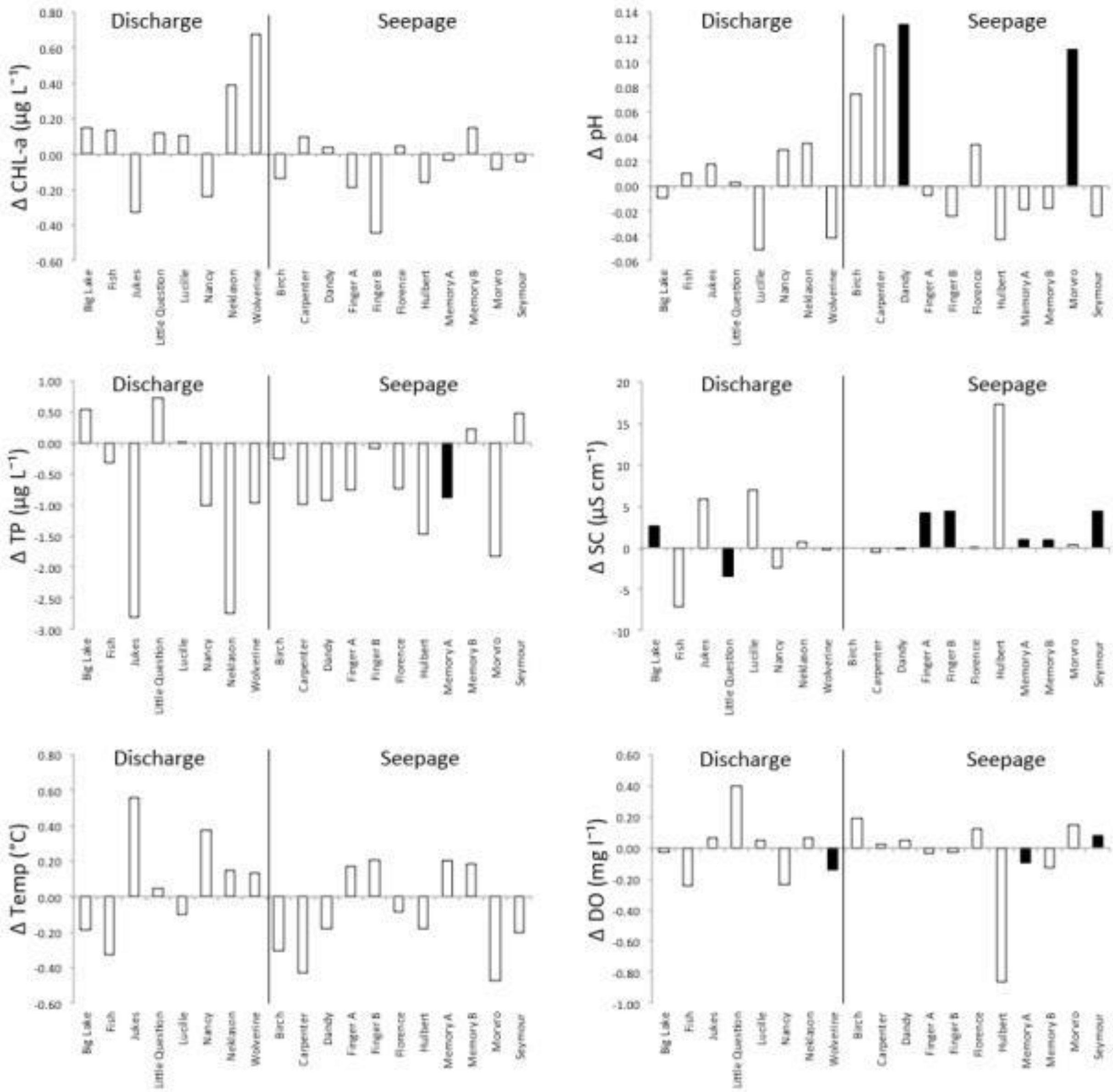

Figure 11. Average change per year calculated with linear regression for 6 water quality parameters in 17 lakes with significant trends in median values designated in black.

3.2.2 Monthly trends

While overall annual trends for individual lakes showed little change in several water quality variables over time, when averaged across lakes by month, May indicated the most change over the last decade. May showed substantial increases from 2001 in SC and CHL-a and the largest decreases in pH and DO (Figure 12). Although trends were small, temperature increased in every month of the year except May, and TP decreased in every month of the year except May.

Specific conductivity increased more than the other water quality parameters accounting for 50% of statistically significant monthly trends (Table 6). Seepage lakes appeared to be more sensitive to change than discharge lakes as there were three times as many significant monthly trends (30 of 102 total possible month and lake combinations showed statistically significant trends) as discharge lakes (10 statistically significant trends). Of the lakes that increased in SC, Big Lake and Finger increased more in the wet season (August-October), while Memory, Juke's and Seymour increased more in May and June when conditions were drier.

Many trends were water quality variable and lake specific. The lakes that had statistically significant decreases in pH also increased in SC during the same month, but lakes with statistically significant increases in pH, increased in CHL-a or temperature. Big Lake became less alkaline in August and September as it increased in SC significantly. Neklason and Carpenter lakes increased in pH in June or July and also increased in CHL-a or temperature.

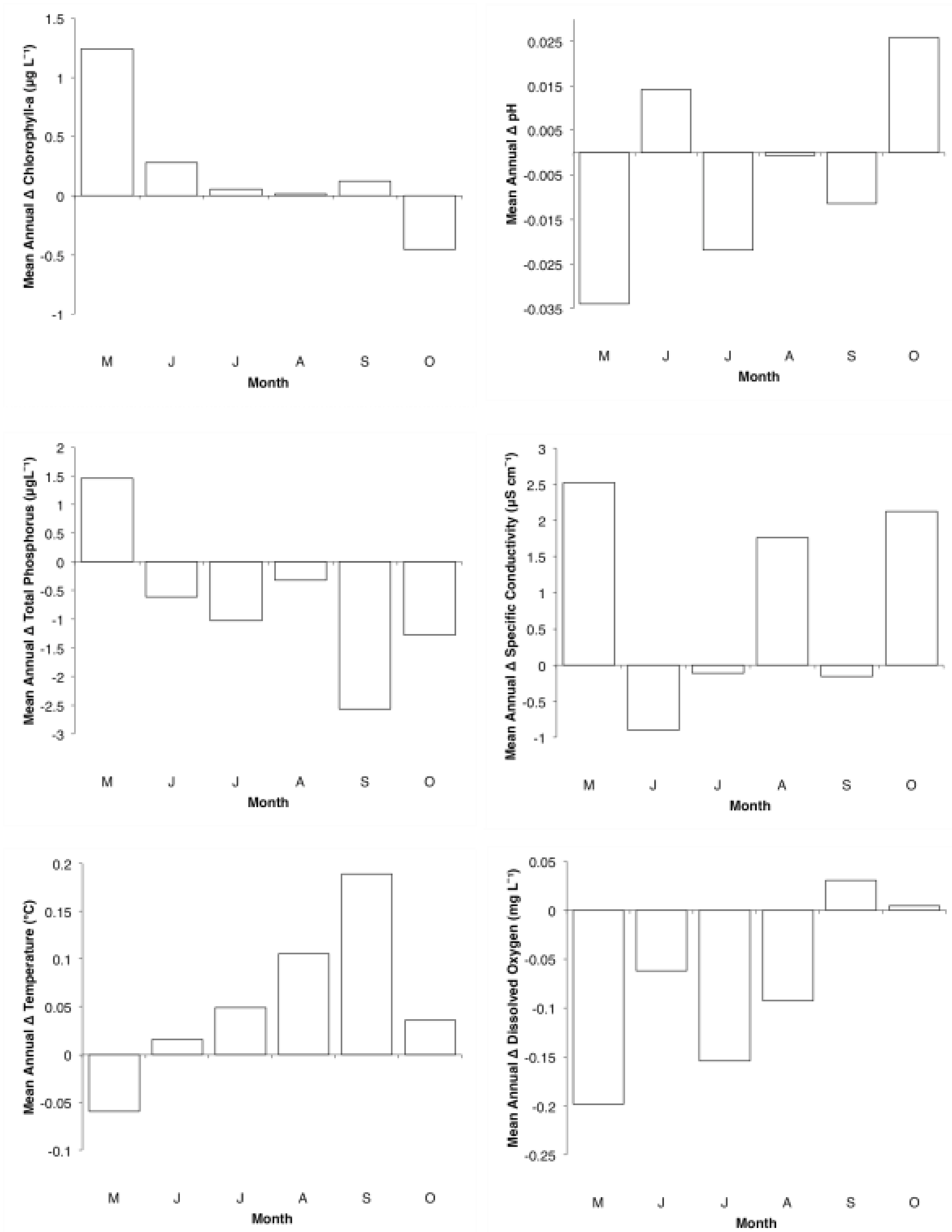

Figure 12. Mean trends for each month from 1998-2014 for lakes in the Mat-Su Valley (n=17) for six water quality parameters measured during the ice-free season; Chlorophyll-a, total phosphorus, temperature, dissolved

oxygen, pH, and specific conductivity. Lakes with fewer than 4 years of water quality parameters measured in any given month were not included in the analysis of monthly trends.

Table 4. Significant trends in water quality averaged for each month measurements were taken from 1998-2014. Discharge and seepage lakes are represented by the average change each year (Trend) determined by linear regression, the water quality variable that changed (WQV), and the number of years (n) measurements were taken for each of the 17 lakes used in the trend analysis. *** = p < 0.001, ** = p < 0.01, * = p < 0.05.

Lake	Hydrology	May			June			July			August			September			October		
		Trend	WQV	n	Trend	WQV	n	Trend	WQV	n	Trend	WQV	n	Trend	WQV	n	Trend	WQV	n
Big Lake	Discharge	-	-	-	**3.00*** **	SC (µS cm^{-1})	8	-	-	-	**4.10** **	SC (µS cm^{-1})	8	-	-	-	-	-	-
	-	-	-	-	-	-	-	-	-	-	**-0.19** **	pH	7	**-0.14***	pH	-	-	-	-
Fish	Discharge	-	-	-	-	-	-	-	-	-	**-0.65***	Temp (°C)	7	-	-	-	-	-	-
Juke's	Discharge	-	-	-	**7.59***	SC (µS cm^{-1})	4	-	-	-	-	-	-	-	-	-	-	-	-
Little Question	Discharge	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Lucille	Discharge	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Nancy	Discharge	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Neklason	Discharge	-	-	-	**0.44***	CHL-a (µg l^{-1})	7	-	-	-	-	-	-	-	-	-	-	-	-
	-	-	-	-	**0.06***	pH	8	**0.07***	pH	7	-	-	-	-	-	-	-	-	-
Wolverine	Discharge	**0.52***	Temp (°C)	5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Birch	Seepage	-	-	-	-	-	-	**-1.64***	TP (µg l^{-1})	3	**0.18***	DO (mg l^{-1})	5	-	-	-	-	-	-
Carpenter	Seepage	-	-	-	**0.30***	pH	5	-	-	-	-	-	-	-	-	-	-	-	-
	-	-	-	-	**0.80***	Temp (°C)	5	-	-	-	-	-	-	-	-	-	-	-	-
Dandy	Seepage	-	-	-	-	-	-	-	-	-	**-0.21***	SC (µS cm^{-1})	7	-	-	-	-	-	-
Finger A	Seepage	**3.13***	SC (µS cm^{-1})	8	-	-	-	-	-	-	**4.04***	SC (µS cm^{-1})	7	**4.68***	SC (µS cm^{-1})	6	**4.37** **	SC (µS cm^{-1})	8
Finger B	Seepage	**3.40***	SC (µS cm^{-1})	8	-	-	-	**3.71***	SC (µS cm^{-1})	7	**5.17***	SC (µS cm^{-1})	7	**4.82***	SC (µS cm^{-1})	6	**4.99** **	SC (µS cm^{-1})	7
Florence	Seepage	-	-	-	-	-	-	**0.19***	CHL-a (µg l^{-1})	7	**-0.30***	Temp (°C)	8	-	-	-	-	-	-
Hulbert	Seepage	-	-	-	**-0.31***	CHL-a (µg l^{-1})	5	-	-	-	-	-	-	-	-	-	-	-	-
Memory A	Seepage	**0.56***	SC (µS cm^{-1})	10	**0.76***	SC (µS cm^{-1})	14	-	-	-	-	-	-	-	-	-	-	-	-
Memory B	Seepage	**0.66***	SC (µS cm^{-1})	10	**0.77** **	SC (µS cm^{-1})	14	-	-	-	**-0.29***	Temp (°C)	12	-	-	-	**0.55***	CHL-a (µg l^{-1})	14
	-	**-1.04***	CHL-a (µg l^{-1})	6	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Morvro	Seepage	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Seymour	Seepage	**6.28***	SC (µS cm^{-1})	11	**4.14*** **	SC (µS cm^{-1})	14	**3.64*** **	SC (µS cm^{-1})	12	**3.11*** **	SC (µS cm^{-1})	13	**3.16*** **	SC (µS cm^{-1})	9	-	-	-
	-	-	-	-	-	-	-	**0.13***	DO (mg l^{-1})	12	-	-	-	-	-	-	-	-	-

3.3 Landscapes around lakes

3.3.1 Land cover

Generally landscapes within 100 meters of lake shorelines ranged from 50% developed land cover to none based on National Land Cover Dataset (NLCD; Figure 2). Lakes averaged 10% developed land cover, which consisted of mostly Low Intensity (5%) and Open Space (4%), and was more prevalent near Palmer and Wasilla in the Core Area (Figure 3). Landscapes were mostly forested (67%), which was mostly deciduous (mean = 16.4 ha), except for Lake Louise, Susitna, and Tyone, which were located in a different lake-district than the other lakes, the Ahtna Lake-District. Wetlands were prevalent in almost all 100 m buffers (7.58 ha), as was open water (4.4 ha). Agricultural, shrub, and barren land covers were minimal, each accounting for less than 1% of total land cover (Table 3).

3.3.2 Geomorphology

Landscapes surrounding lakes were steeper near the Matanuska River (10°, n=6) and in the Core Area (7°, n=12), and flatter near the Susitna River (4.5°, n=12). Soil particle sizes and texture differed among the lake's 100 m buffers depending on major watershed (Susitna vs. Matanuska). Coarse silty over sandy or sandy skeletal soils (Knik Silt Loam) were more prevalent in the Matanuska River Watershed and Core Area near Palmer and Wasilla, while medial and fine loamy soils were more prevalent closer to the Susitna River (Figure 4, 5).

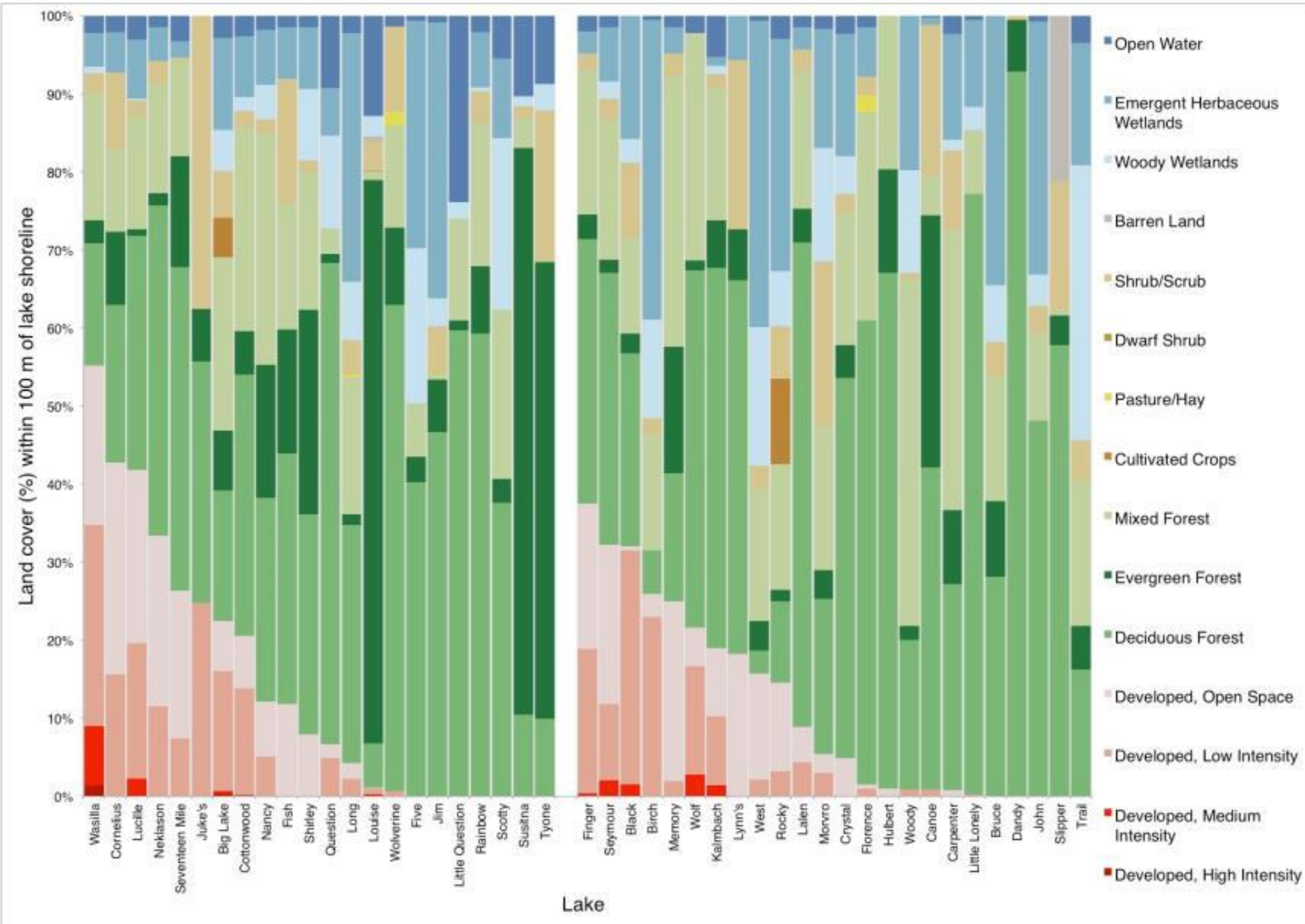

Figure 2. Percent of National Land Cover Dataset coverage within 100 meters of discharge lake shorelines (left; n=22) and seepage lakes (right; n=22) arranged from the highest percentage of development to the lowest development.

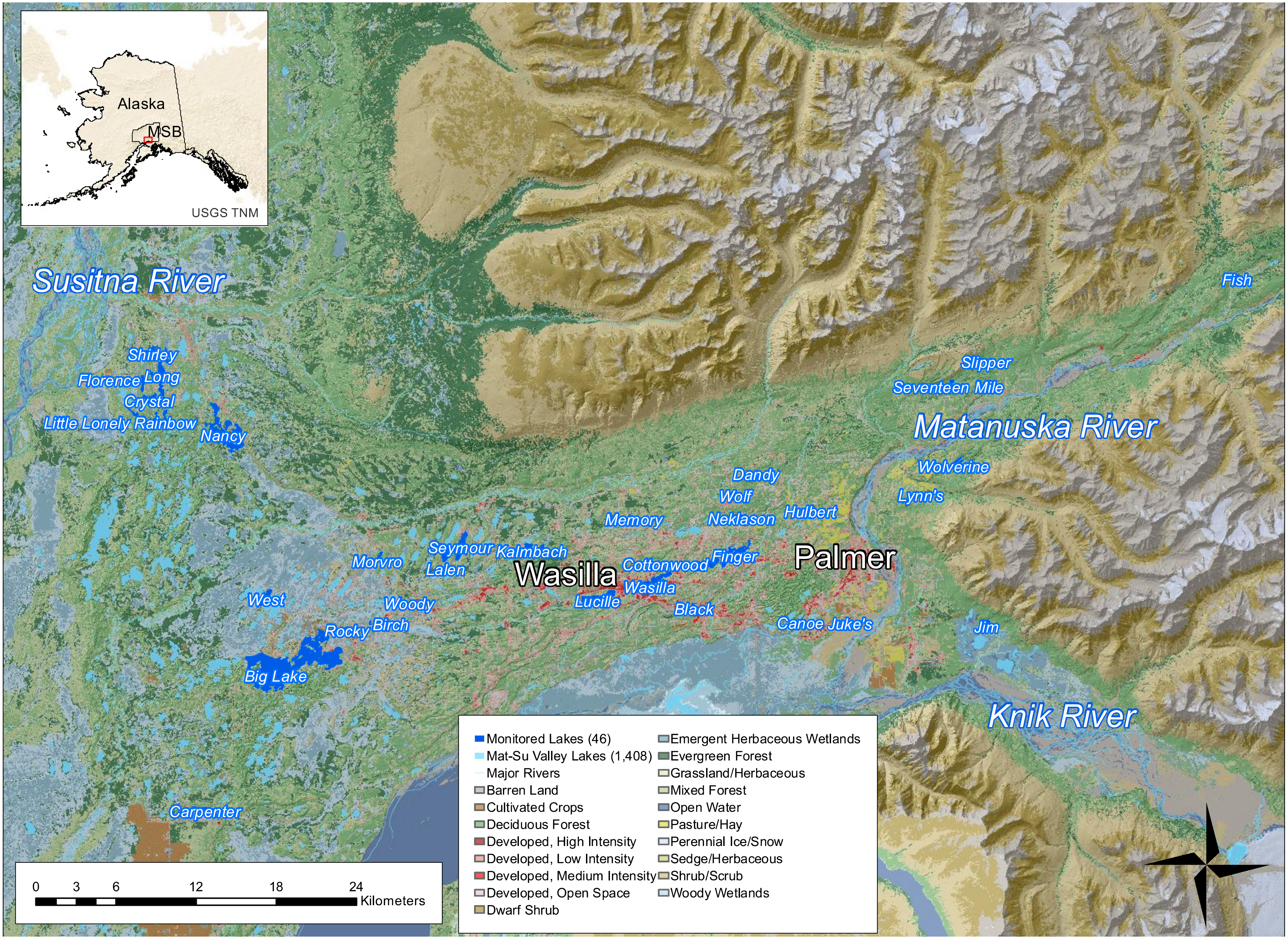

Figure 3. National Land Cover Data and lakes monitored in the Mat-Su Valley.

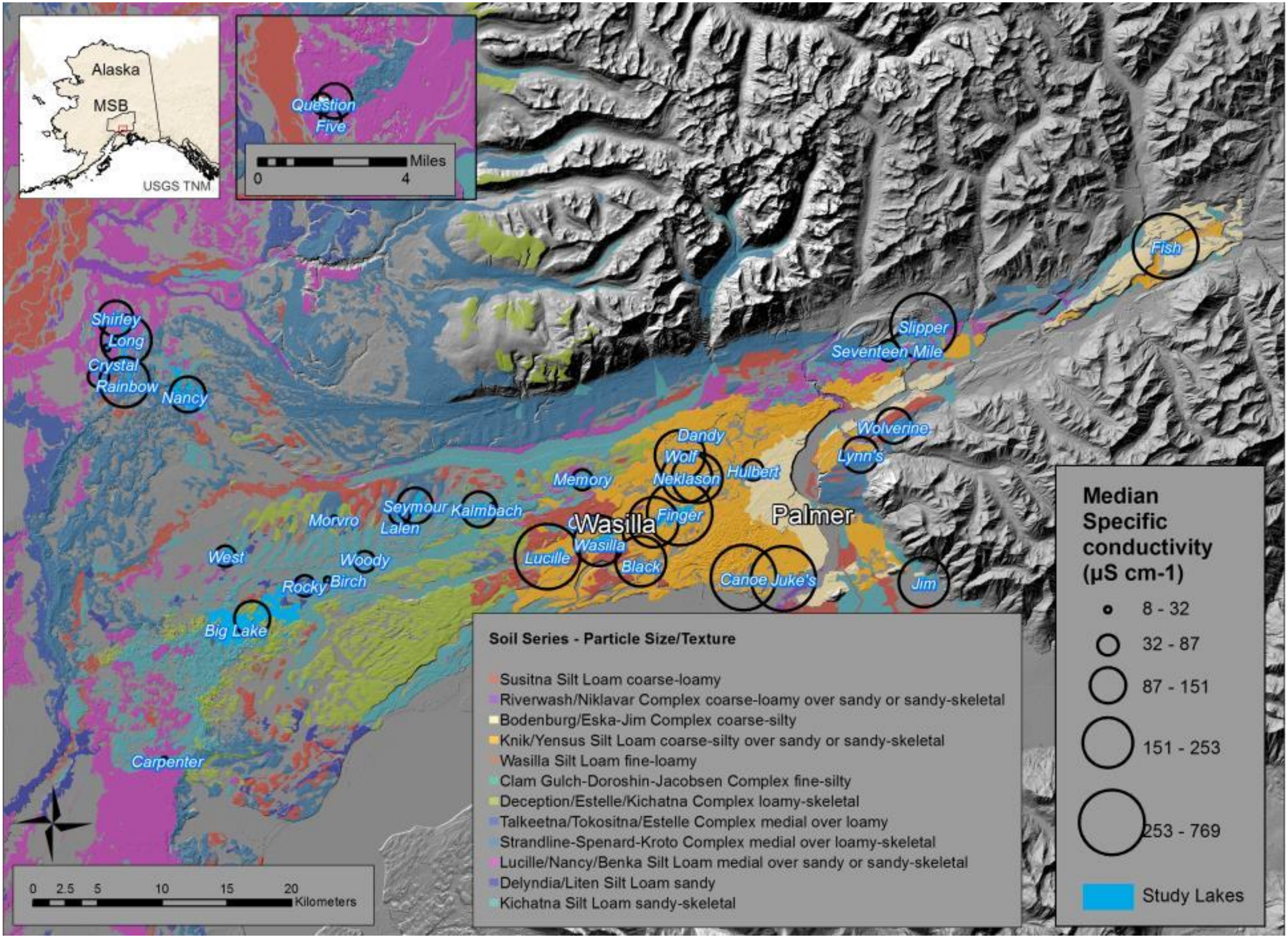

Figure 4. Mapped soil series and particle size and median specific conductivity represented by circle size for lakes in the Mat-Su Valley.

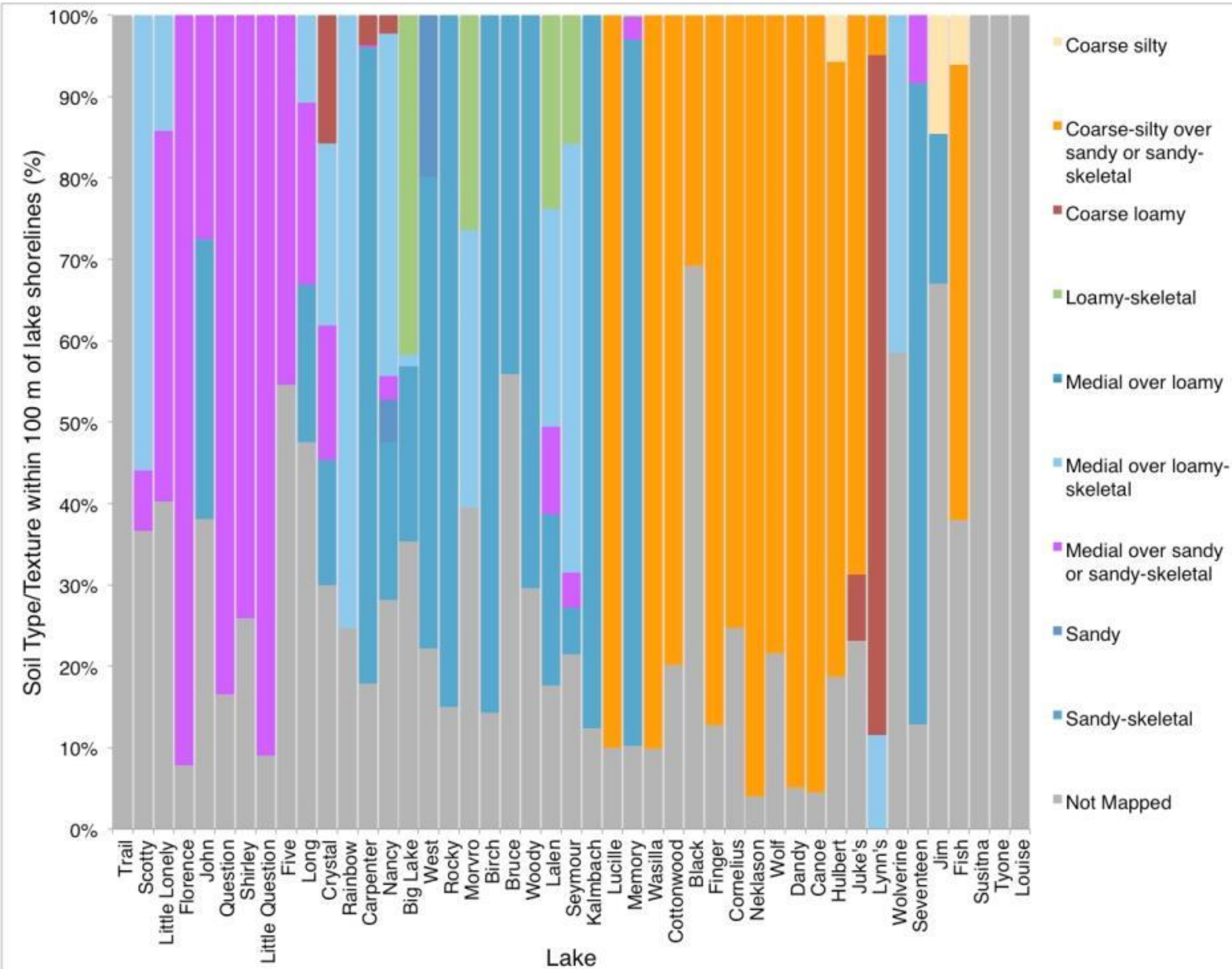

Figure 5. Soil texture/taxonomic class as a percentage of total mapped soil area within watersheds for 42 lakes in the Mat-Su Valley arranged from west to east.

Table 5. Summary of buffer landscape variables within 100 m of the shorelines of 46 lakes in the Mat-Su Borough.

Landscape Variable	n	Mean	Median	SD	Min	Max
100 m Buffer Slope (degree)	43[a]	6	6	4	2	17
100 m Buffer Area (ha)	46	291	37	1016	1	5990
Buildings (number of structures)	46	61	22	122	0	755
Total Developed Land cover (ha)	46	8	2	16	0	80
Total Developed Land cover (%)	46	13	8	15	0	55
Developed, High Intensity (%)	46	0	0	0	0	1
Developed, Medium Intensity (%)	46	0	0	1	0	8
Developed, Low Intensity (%)	46	6	2	8	0	30
Developed, Open Space (%)	46	6	2	8	0	27
Total Agriculture Land cover (ha)	46	0	0	3	0	18
Total Agriculture Land cover (%)	46	0	0	2	0	11
Pasture/Hay (%)	46	0	0	0	0	2
Cultivated Cropland (%)	46	0	0	2	0	11
Total Forest Land cover (ha)	46	50	22	105	3	604
Total Forest Land cover (%)		62	64	19	19	99
Deciduous Forest (%)	46	36	33	20	3	93
Evergreen Forest (%)	46	11	6	17	0	73
Mixed Forest (%)	46	15	16	10	0	44
Total Shrub Land cover (ha)	46	3	1	7	0	37
Total Shrub Land cover (%)	46	6	3	8	0	38
Open Water (ha)	46	4	1	14	0	72
Total Wetland Land cover (ha)	46	8	5	12	0	61
Total Wetland Land cover (%)	46	16	9	17	0	57
Emergent Herbaceous Wetlands (%)	46	11	7	12	0	39
Woody Wetlands (%)	46	5	2	7	0	35
Barren Land (ha)	46	0	0	1	0	3
Barren Land (%)	46	0	0	0	0	1
Coarse Loamy Soil (ha)	42[b]	1	0	4	0	25
Coarse Silty Soil (ha)	42[b]	0	0	1	0	5
Coarse Silty over Sandy/Skeletal	42[b]	13	0	29	0	135

(ha)						
Loamy Skeletal (ha)	42[b]	4	0	23	0	153
Medial over Loamy	42[b]	0	0	2	0	11
Medial over Loamy Skeletal (ha)	42[b]	7	0	18	0	108
Medial over Sandy/Skeletal (ha)	42[b]	4	0	8	0	30
Sandy (ha)	42[b]	1	0	2	0	13
Sandy Skeletal (ha)	42[b]	10	0	17	0	79

[a] Three lakes did not have available digital elevation model data available for surrounding landscape
[b] Four lakes did not have soil data mapped in the surrounding landscape.

3.4 Lake Water Quality – Landscape Relationships

Among water quality variables, geomorphologic variables, and hydrology, the strongest relationships were between SC, Knik vs. non-Knik Soils, hill slope, and discharge vs. seepage hydrology. Among land cover variables wetlands and developed land cover area were most strongly correlated with SC. In addition, lake temperature was moderately correlated with hill slope and lake depth as well as SC.

3.4.1 Geomorphology and Hydrology

Warm lakes were generally low in SC while cold lakes were high in SC ($p < 0.01$, $r = 0.41$, $n = 46$) but temperature and SC both depended on the geomorphology and hydrology of the Susitna or Matanuska Watersheds (Figure 6). Seepage lakes, shallow lakes, lakes with low angle slopes within 100-m of the shoreline, and lakes in the Susitna River Watershed were warmer and lower in SC. Discharge lakes, deep lakes, lakes with steeper slopes within 100-m of the shoreline, and lakes in the Matanuska River Watershed were colder and higher in SC. These relationships among variables are apparent in ordination.

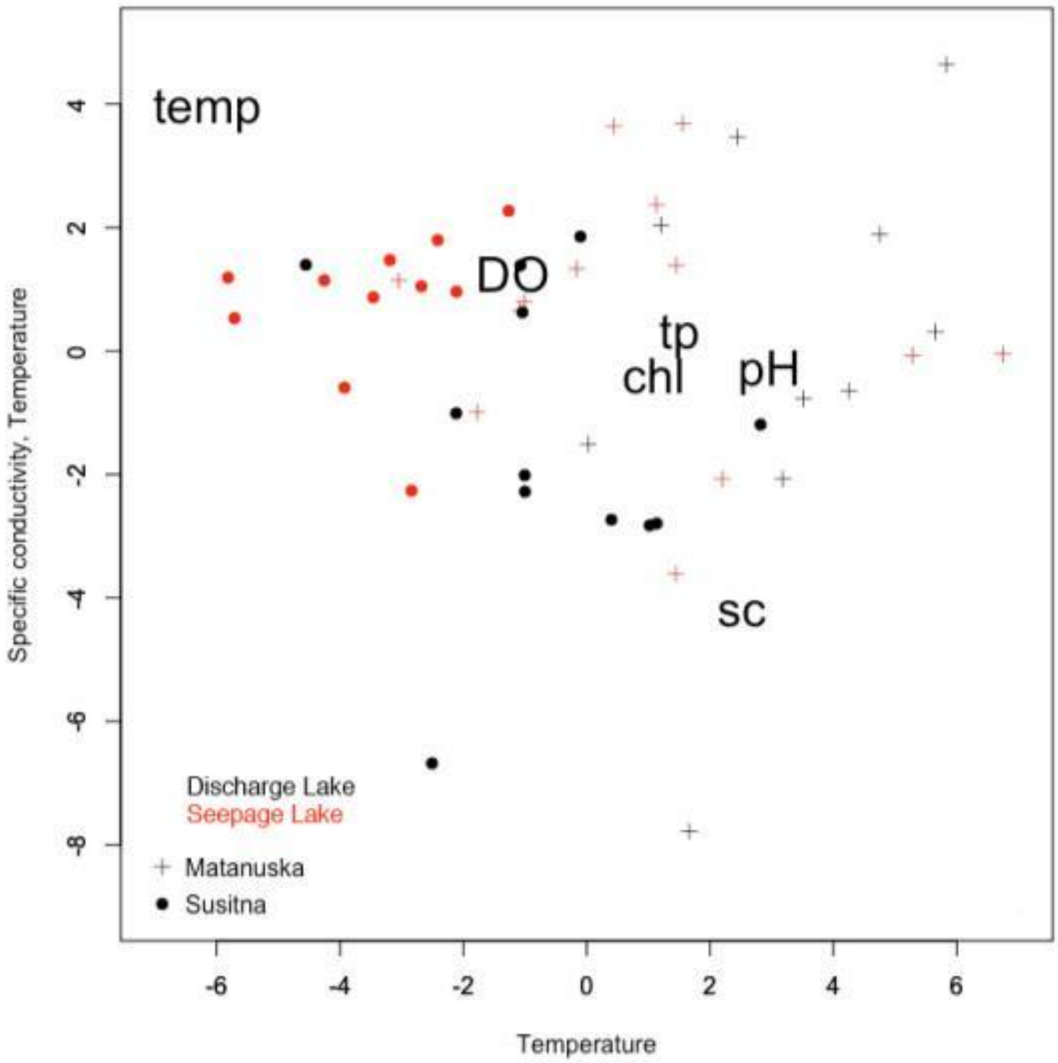

Figure 6. A Non-metric Multidimensional Scaling (NMS) ordination of lakes (n = 46) based on 4 median lake water quality variables, specific conductivity (SC), pH, water temperature (temp), and dissolved oxygen (DO). Symbols denote lakes closer in the Matanuska River Watershed (n= 23, +) or the Susitna River Watershed (n = 23,•). Colors denote the hydrology of a lake, discharge (n = 22, black) or seepage (n = 24, red).

Using Non-metric Multidimensional Scaling (NMS) ordination showed that lakes clustered by major watershed (Susitna vs. Matanuska) and by hydrology (seepage vs. discharge) along axes defined by temperature and conductivity. Axis 1 of the NMS and axis 2 correlated best with lake specific conductivity and temperature, thus were named the SC and Temperature Axes. Lakes with high SC were more positive in the SC axis, and lakes with high temperatures were more positive in the Temperature Axis. Seepage lakes, which were warmer and had lower SC, were more positive in the Temperature Axis and negative in the SC Axis in the NMS. Discharge lakes were more varied in their water quality but generally had higher SC, and lower temperatures than seepage lakes, thus were positive in the SC Axis, and negative in the Temperature Axis in the NMS. Lakes in the Core Area appeared to be less related in water quality than other lakes in other regions, although lakes in the Cottonwood Creek watershed were very similar in water quality to each other, with high pH and high SC. The NMS identified two lakes that were relative outliers, Fish and Little Question lakes. They were much higher in median SC with much lower median temperatures compared to other lakes.

3.4.2 Soils and Land cover

High lake SC was correlated with the areal extent of Knik Silt Loam (KSL) soils (coarse silty over sandy or sandy skeletal loess) within a lake's 100 m buffer ($r = 0.54$, $p < 0.01$, $n = 42$) and with steep slopes ($r = 0.44$, $p < 0.01$, $n = 42$; Figure 6). Developed land cover and wetlands were the most important land cover correlates with SC (Table 4). Ordination using Principal Components Analysis (PCA) (Figure 7) indicated correlations with SC (and with TP) for wetlands ($r < 0$) and for developed land cover ($r > 0$). A second axis showed correlations between water temperature, dissolved oxygen, hillslope, and maximum depth as with NMS ordination. Lakes in the Cottonwood Creek Watershed in the Core Area of the Mat-Su Valley (numbered 7, 8, 11, 25, 30, 42 in Figure 7) were higher in SC, TP, CHL-a, pH, had coarse silty soils, and had more development than the rest of the lakes. Lakes in the Susitna Watershed were warmer, were less productive, less alkaline, shallower, and had more wetland cover.

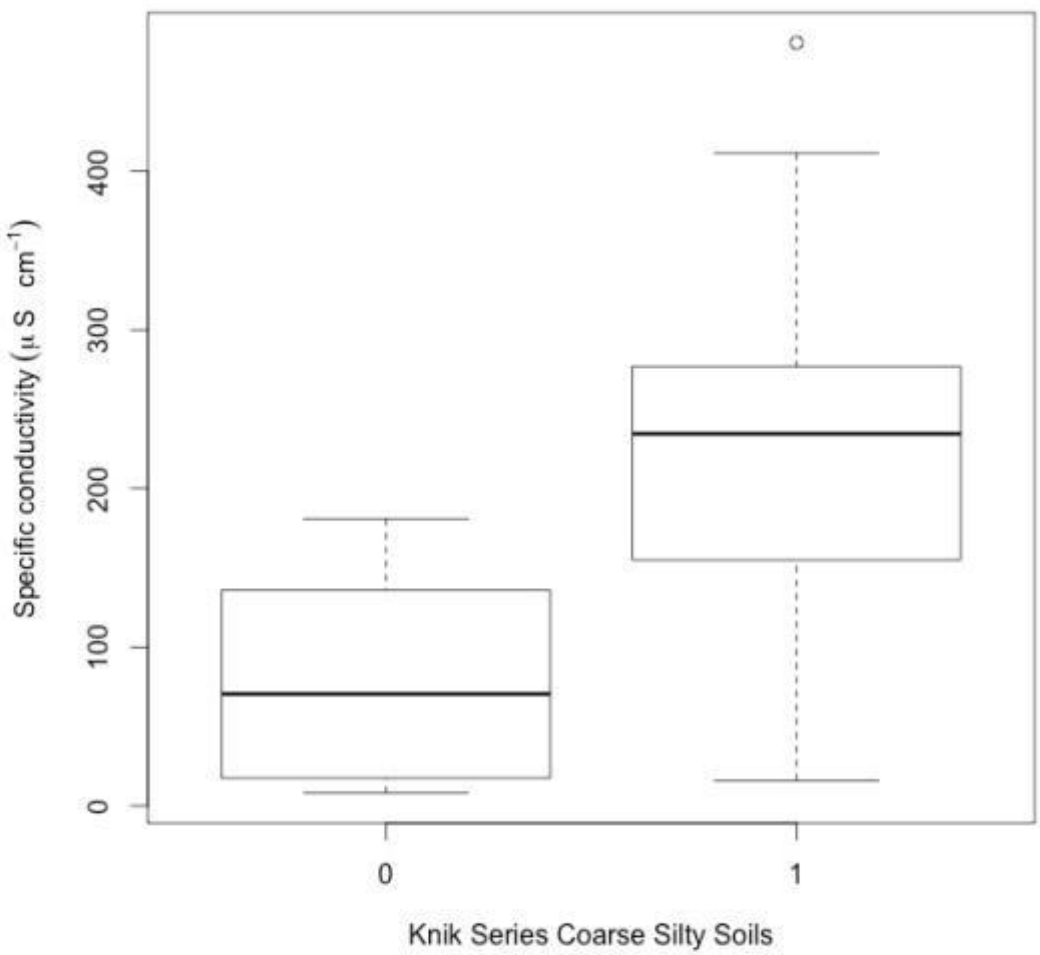

Figure 7. Median specific conductivity (SC) and the presence of Knik Silt Loam coarse silty over sandy skeletal soils within 100-m of lake shorelines (n=42).

Table 6. Correlation matrix of median lake water quality variables and environmental variables using Pearson's correlation for Total Phosphorus (TP), chlorophyll-a (CHL-a), Temperature, dissolved oxygen (DO), specific conductivity (SC), and pH. p < 0.001 = ***, p < 0.01 = **, p < 0.05 = *

	TP (µg L^{-1})	CHL-a (µg L^{-1})	SC (µS cm-1)	pH	DO (mg L-1)	Temperature (°C)
TP (µg L^{-1})						
CHL-a (µg L^{-1})	0.41**					
SC (µS cm-1)	0.19	0.16				
pH (SU)	0.23	0.19	0.71***			
DO (mg L-1)	-0.03	-0.03	-0.05	0.29		
Temperature (°C)	-0.07	-0.11	-0.41**	0.02	0.44**	
Max depth (m)	-0.28	0.05	0.2	0.02	-0.25	-0.59***
Perimeter (km)	-0.36*	0.17	0.24	0.15	0.1	-0.22
Surface area	-0.33*	0.15	0.17	0.15	0.07	-0.14

100-m Buffer Hillslope (degree)	0.14	0.21	0.44**	0.23	-0.19	-0.62***
Developed land cover (ha)	-0.07	0.4**	0.32*	0.34*	0.14	-0.2
Forest Cover (ha)	-0.31*	0.14	0.23	0.12	0.03	-0.25
Wetland (ha)	-0.6***	0.07	-0.26	-0.25	0.11	0.19
Agriculture (ha)	-0.66***[a]	-0.22	0.03	0.05	0.06	0.06
Shrublands (ha)	-0.36*	-0.14	0.37*	0.29	0.13	-0.19
Coarse Loamy (ha)	-0.13	0.05	0.18	0.05	0.04	-0.05
Coarse Silty (ha)	0.01	-0.07	0.21	0.19	-0.16	-0.07
(KYSL) Coarse Silty over Sandy/Skeletal (ha)	0.4*	0.45**	0.54***	0.52***	0.1	-0.29
Loamy Skeletal (ha)	-0.53***	-0.21	-0.01	0.11	0.07	0.09
Medial over Loamy (ha)	-	-	-0.25	-0.11	-0.02	0.16
Medial over Loamy Skeletal (ha)	-0.28	0.07	-0.07	-0.12	0.01	0.06
Medial over Sandy/Skeletal (ha)	-0.34*	-0.08	-0.24	-0.43**	-0.23	-0.15
Sandy (ha)	-0.55***	-0.23	0.06	0.13	0.05	0.08
Sandy Skeletal (ha)	-0.37*	-0.06	-0.19	-0.15	0.08	0.17
Barren Land (ha)	0.09	-0.06	0.25	0.11	-0.01	-0.14
Cultivated Cropland (ha)	-0.64***[a]	-0.22	0.05	0.06	0.06	0.02
Deciduous Forest (ha)	-0.3*	0.24	0.22	0.16	0.01	-0.25
Developed High Intensity (ha)	-0.27	0.23	0.15	0.11	0.07	-0.1
Developed Low Intensity (ha)	-0.1	0.41**	0.32*	0.38**	0.19	-0.18
Developed Medium Intensity (ha)	-0.04	0.29	0.25	0.28	0.11	-0.05
Developed Open Space (ha)	-0.04	0.46**	0.27	0.24	0.07	-0.21
Dwarf Shrub (ha)	-0.03	-0.09	0.07	0.01	0.04	-0.22
Emergent Herbaceous Wetland (ha)	-0.54***	0.18	-0.29	-0.22	0.1	0.3*
Evergreen Forest (ha)	-0.21	-0.05	0.32*	0.11	0.05	-0.34*
Mixed Forest (ha)	-0.34*	0.32*	0	-0.03	-0.02	0.03
Open Water (ha)	-0.28	0.02	0.28	0.1	-0.07	-0.34*
Pasture/Hay (ha)	-0.14	0	-0.05	-0.03	0.03	0.13
Scrub/Shrub (ha)	-0.36*	-0.14	0.37*	0.29	0.13	-0.19
Woody Wetlands (ha)	-0.55***	-0.09	-0.21	-0.29	0.06	0.05
Buildings	-0.35*	0.25	0.33*	0.3*	0.14	-0.1

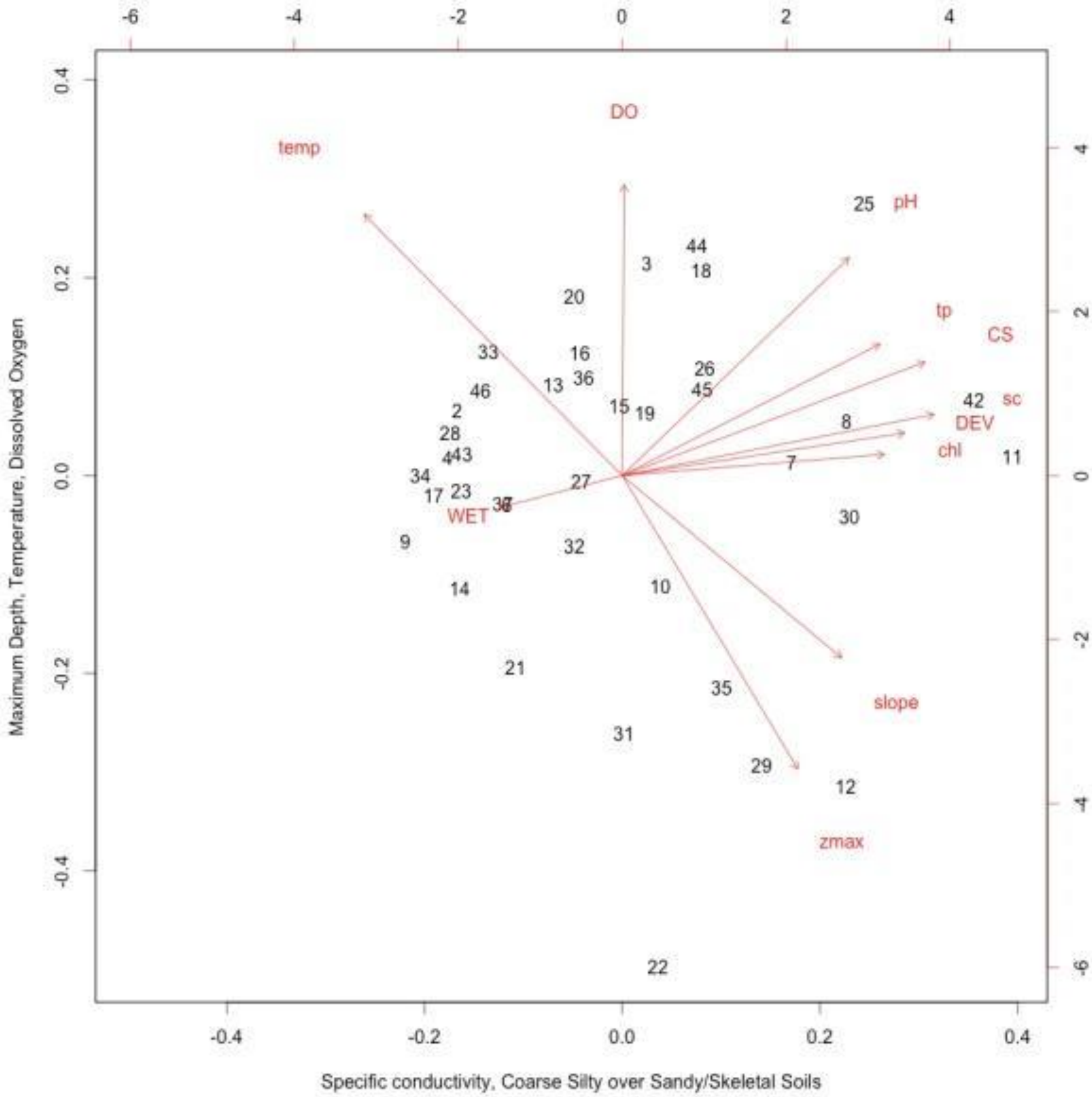

Figure 7. A Principal Component Analysis (PCA) ordination of lakes (n = 39) based on 6 median lake water quality variables, specific conductivity (SC), pH, water temperature (temp), and dissolved oxygen (DO) and 5 variables with the highest correlation with water quality variables. Development (DEV), Maximum depth (zmax), hillslope within 100-m of lake shorelines (slope), wetlands (WET), and coarse silty over sandy or sandy skeletal soils (CS) were the most important variables correlated with median water quality. See Table … for lake numbers.

3.4.3 General Model for Specific Conductivity

Specific conductivity (SC) was the first principle component in the PCA, explained a major axis in NMS, and was well correlated with multiple water quality parameters and environmental variables. Therefore a general linear model of SC as a function of environmental variables confirmed the importance of soils and slope (Equation 1). The presence of Knik Silt Loam series soils, which is coarse silty over sandy or sandy skeletal soils mostly in the Matanuska Watershed and the Core Area, and steeper slopes explained most of the variation in SC among lakes ($r^2 = 0.72$, $p < 0.001$, $n = 41$). Residuals suggested a good fit as they were mostly randomly distributed about zero when plotted against predicted SC (Figure 8).

Equation 1. $$SC = 14.8 + (170.4 * (Knik\ Silt\ Loam\ Coarse\ Silty\ Soils)) + (11.8 * (100\ m\ buffer\ hillslope))$$

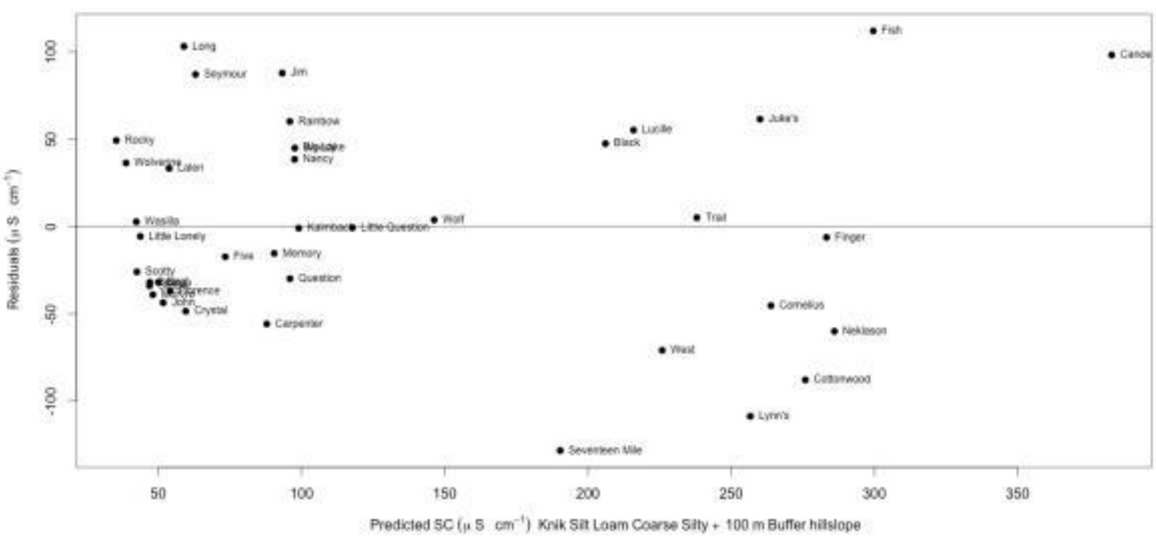

Figure 8. Residual plot of regression model explaining specific conductivity (SC) in 41 lakes in the Matanuska-Susitna Valley where the presence of Knik Silt Loam series soils and steep slopes best explain median SC.

3.5 Temporal Trends

3.5.1 Land cover change

Most land cover change from 2001 to 2011 at both the watershed (Figure 9) and lake buffer (Figure 10) scales was due to increased development. Five of the 15 study watersheds delineated with Terrain works NetMap had land cover change from 2001 to 2011 due to increase in development and accounted for 98% (182.3 ha) of all land cover change. One watershed had emergent herbaceous wetlands change to woody wetlands, one watershed had emergent herbaceous wetlands convert to open water, and eight watersheds had no land cover change. Deciduous, mixed, and coniferous forest loss to development accounted for 73% (135.6 ha) of all development. Of the 15% increase in development across the 5 watersheds, nearly all was Open Space Development (79 ha) and Low Intensity Development (87 ha).

Although watersheds delineated with Terrain works NetMap and 100-m buffers did not differ significantly in slope or land cover composition, they differed in land cover change from 2001 to 2011. Within 100 meters of the 15 lake shorelines only five had land cover change. Three experienced new development, and 87% of new development was developed as Open Space (10.8 ha) and Low Intensity Development (10 ha). There was no High Intensity Development; Medium Intensity Development accounted for only 1.6 ha (Figure 10).

3.5.2 Temperature and precipitation change

During 2001-2011, temperatures were warmer on average across all seven weather stations, but precipitation was higher at some stations and lower at others than 30-year averages (Table 5). Across the Mat-Su Valley, mean annual temperatures were on average 0.57 °C warmer and as much as 1.4°C warmer than 30 year averages. Precipitation was higher than average when averaging total annual precipitation from all weather stations (n=7), but Talkeetna Airport experienced 60 mm less than average yet the weather station at White’s Crossing experienced 53 mm more than average.

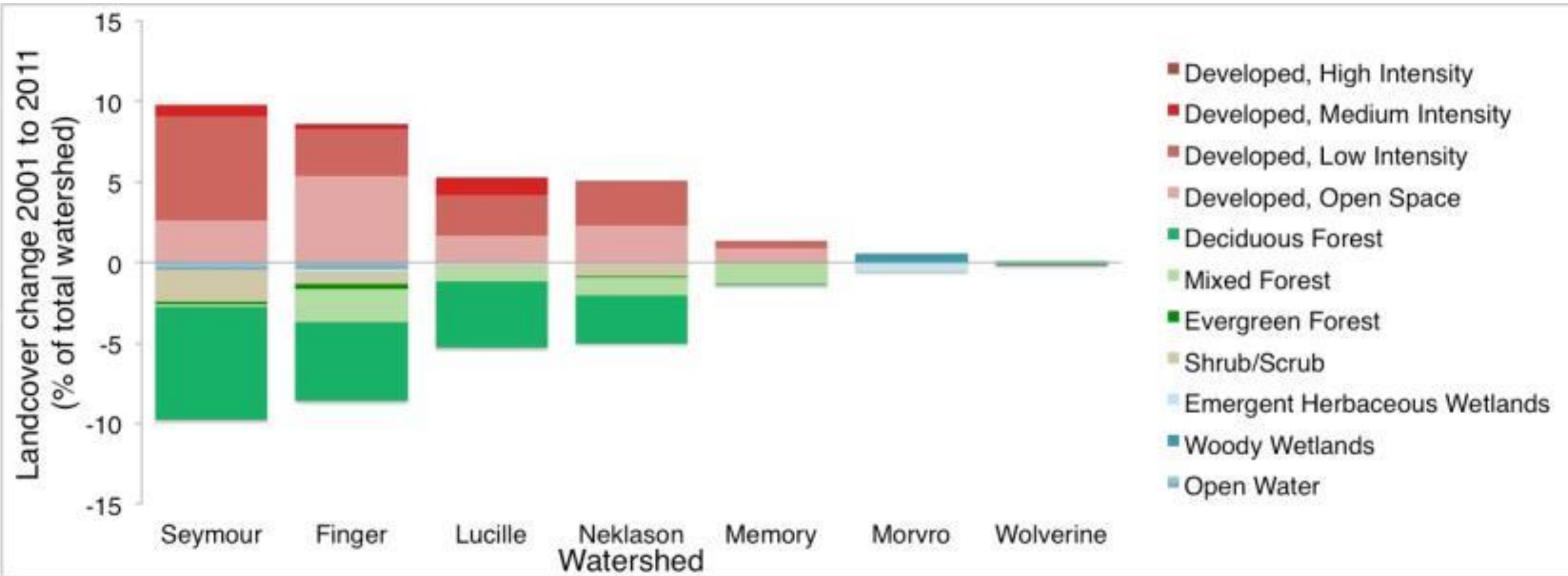

Figure 9. Land cover change from 2001 to 2011 as a percentage of total watershed area in the seven watersheds that had conversion of land within their watershe

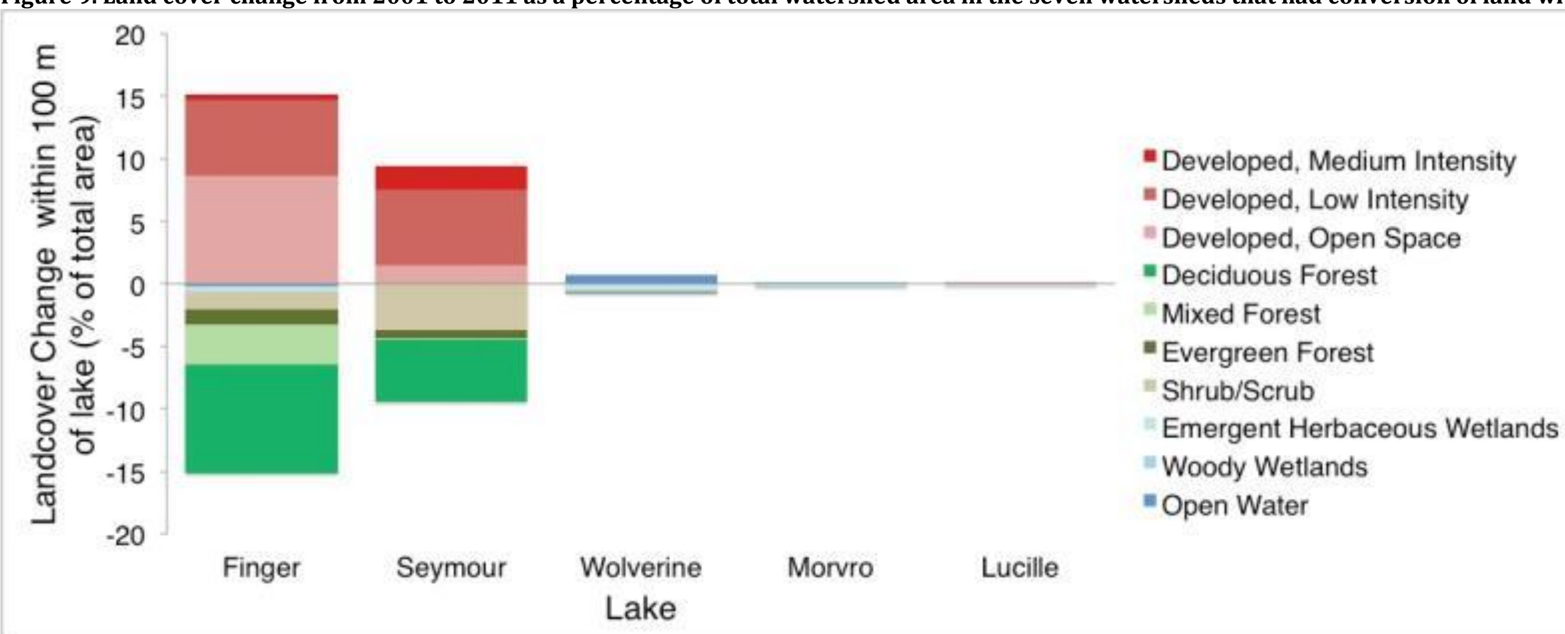

Figure 10. Land cover change from 2001 to 2011 as a percentage of total watershed area in the five watersheds that had conversion of land within their 100-m buffers

Table 6. Summary statistics from seven weather stations in the Mat-Su Valley in close proximity to monitored lakes

Weather Station	Total Record	n (1981-2010)	Total Annual Precipitation (mm) 1981-2010	Total Annual Precipitation (mm) 2001-2011	Precipitation difference from average (mm)	Mean Temperature (°C) 1981-2010	Mean Temperature (°C) 2001-2011	Temperature difference from average (°C)	Latitude (DD)	Longitude (DD)	Elevation (m)
PALMER JOB CORPS, AK US	1948-2015	19	408	445	37.10	2.83	4.24	1.41	61.588800	-149.099200	65.8
TALKEETNA AIRPORT, AK US	1922-2015	29	716	656	-59.82	1.84	2.56	0.72	62.320000	-150.095000	106.7
SUTTON 1 W, AK US	1978-2011	28	469	488	19.21	2.31	2.69	0.38	61.716670	-148.883330	167.6
ANDERSON LAKE, AK US	1971-2010	20	479	443	-36.78	2.76	3.40	0.64	61.624440	-149.339720	143.3
WHITES CROSSING, AK US	1970-2009	25	573	627	53.33	0.32	0.72	0.40	61.706670	-149.997780	92.7
LAZY MOUNTAIN, AK US	1984-2015	22	474	452	-22.21	1.67	1.87	0.20	61.629440	-149.029720	223.4
MATANUSKA AG. EXP. STATION, AK US	1922-2015	28	378	382	3.89	2.41	2.66	0.25	61.566390	-149.254170	52.4

3.6 Land cover change, climate change, and trends in specific conductivity

Because specific conductivity (SC) was the water quality variable that changed most significantly in lakes in the Mat-Su Valley, and because it is well correlated with some other measures of water quality, it was used to test the effect of environmental variables on water quality (Equation 2). Overall, SC increased with more new development, more precipitation, and shallower lakes. During the period lakes were monitored, specific conductivity increased more in lakes that had more forest cover change to developed land cover in the watershed, that were shallower, and that had higher than average total annual precipitation ($p < 0.001$, $R^2 = 0.87$, $n = 15$, AIC = 136.8). Lakes with more than 20 ha of new developed land cover from 2001 to 2011 within their watershed had significantly higher increases in SC than lakes with between 2 and 20 ha of new developed land cover. Lakes with more than 2 ha of new development had higher increases in SC than lakes with no new developed land cover (Figure 14). Hulbert Pond was not included in this model because it lowered the r^2 to 0.74 and raised the AIC to 159.6, suggesting there is an alternative effect to SC change despite the random and normally distributed residuals (Figure 13).

Equation 2.

$$\Delta SC = 16.4 + (33.6 * (2\ ha < \Delta\ Developed\ Land\ Cover < 20 ha) + \big(83.1 * (\Delta\ Developed\ Land\ Cover > 20\ ha)\big) + \big(0.44 * (\Delta Total\ annual\ precipitation)\big) - (3.5 * \big((maximum\ depth)\big))$$

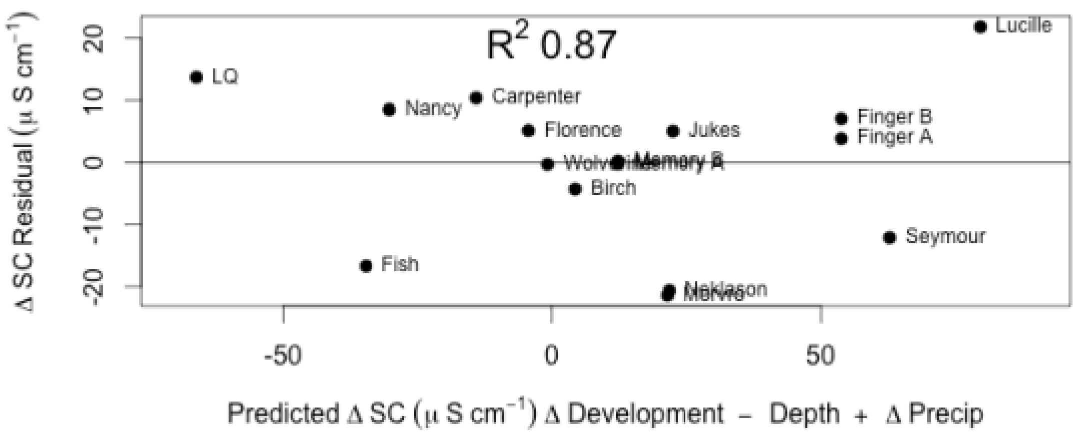

Figure 13. Residual and predicted specific conductivity (SC) change over time from a model using linear regression ($r^2 = 0.87$) and independent environmental variables: (1) 3 levels of watershed scale development increase, (2) maximum lake depth, and (3) Δ precipitation.

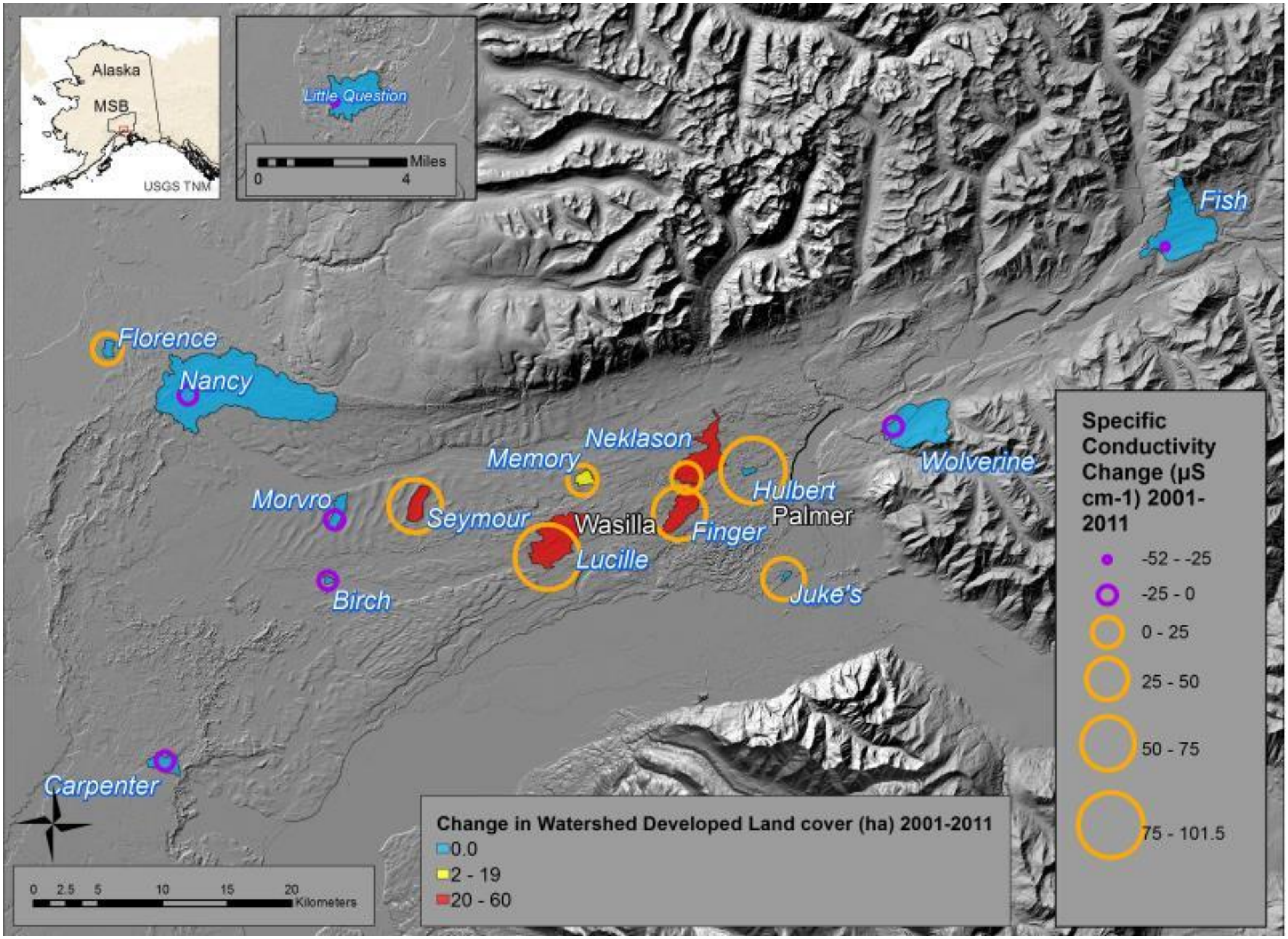

Figure 14. Specific conductivity change in lakes (n=15) and increase in developed land cover in the Mat-Su Valley.

4 Discussion

This study considered water quality and landscape variables among 46 lakes over 15 years in the Matanuska-Susitna Valley of Southcentral Alaska, the fastest growing region in the State. The landscapes surrounding lakes in the Mat-Su Valley are mostly rural and forested but development has increased substantially in some areas, mostly near Wasilla in the form of residential home building. Concurrently, specific conductivity increased significantly in the lakes with the most new development in their watersheds between 2001 and 2011 and total phosphorus decreased when averaged across lakes. Lakes that were shallower were more susceptible to changes in specific conductivity; increased precipitation in some areas increased SC in lakes with less than 20 ha of new development between 2001 and 2011. In a relatively few number of lakes, pH decreased while specific conductivity increased and total phosphorus decreased in almost all lakes.

Understanding water quality change in the context of anthropogenic disturbance in the sub-arctic, where average temperatures have increased significantly, requires careful analysis and identification of variables that may be important drivers of change. For example, untangling the complexities of lake-landscape relationships as warmer temperatures decrease areal extent of some lakes (Rover 2014) and accelerate the phenology of zooplankton (Carter 2012) are important for anticipating potential ecosystem change in a changing climate.

Water quality in most lakes has not changed substantially enough to warrant intervention; however, monitoring should continue. While some lakes did change substantially over time, with total phosphorus decreasing on average and SC increasing in most lakes, temperature, DO, CHL-a, and pH, did not change in a statistically significant sense in most lakes.

This study identified four potentially important environmental variables that may determine background values for water quality. The variables included lake depth, the steepness of the slope within 100 m of the lake, the presence of Knik Silt Loam, and the dominant source of water inputs into lakes either by streams or groundwater. Land cover had some correlation to lake water quality but was a minor component; only the amount of wetlands and the amount of development correlated with total phosphorus (TP) and specific conductivity (SC), respectively.

While the findings of this research suggest land cover change is a possible driver of increases in the dissolved ion input in some lakes, the strong relationship between high specific conductivity and Knik Silt Loam -- loess deposits primarily from the Matanuska Valley (Muhs 2004, Muhs 2016) -- must be further examined as an alternative hypothesis. The effect of wind from the Matanuska and Knik valleys was not evaluated in this study but is likely important in the interaction between land use and climate effects. It is possible that wind exacerbates the effect of forest removal around these lakes and contributes to further erosion and deposition of material in the lakes. Further inquiry of the effects of wind and runoff may also shed some light on the observed decrease in total phosphorus across the Mat-Su Valley. Phosphorus readily binds to many dissolved ions in the water column and may have been less available for sampling due to an increase in specific conductivity, a measure of dissolved ions. Regardless of the primary driver of water quality change, lakes in the area where development is the most intense and the wind is strongest will need careful monitoring and to ensure the amount of dissolved solids do not exceed water quality standards for aquatic life.

4.1 Water quality is generally good considering impervious surface threshold

Lake water quality was generally good with little change over time. The small amount of change that was observed in lakes were isolated to increases in specific conductivity in lakes with more new development, an overall decrease in total phosphorus, and a few lake specific changes. The changes observed likely reflect wind, land cover changes, hydrology, and depended on lake depth. The climate of southcentral Alaska is warming with less winter snowfall and more rain, affecting the hydrology, which is a very important determinate of lake water quality in the Mat-Su Valley. Land cover change and the subsequent increase of impervious surfaces have been shown to be a contributing factor affecting water quality all over the world. Land cover change in the Mat-Su is important because there has been concern over the effects of development on lake water quality since the 1980's. Brabec et al (2002) suggested an impervious surface thresholds of 10% within a watershed before water quality may become impaired. But overall, lake water quality responded to anthropogenic and climatic change with little overall change in water quality and expected seasonal variability.

The lack of trends present in lakes is an important result considering the dramatic rise in population and the warmer climate that the Mat-Su Valley experienced from 2001-2011. This suggests that most of the Mat-Su Valley has not reached the threshold for significant changes to water quality and localized deforestation is not widespread enough to have a strong effect.

This study establishes a baseline in the Mat-Su Valley to refer to as population increases and the climate warms. Projecting into the future, the trends identified during this study may not be significant over the next century of lake, landscape, and climate change. These trends occurred over a relatively short period of time. The Mat-Su Valley will undoubtedly continue to grow in population but maintain a relatively rural setting with a select few locations becoming more urbanized. Many lakes will have new development and experience more use from recreation. Additionally, lakes that will likely continue as susceptible to changes in land cover and climate include Finger, Hulbert, Juke's, Lucille, Seymour, Memory, and lakes of similar depth and lakes nearby because of their location in the wind paths and substantial forest loss to development. Management efforts to protect water quality must focus on these lakes with scientific evidence gathered from the lake monitoring program.

As part of this study, an evaluation of the quality of the data was conducted to verify its validity and to identify potential problems from sampling that could affect analyses. A high standard is expected for water quality sampling; each citizen-scientist participates in an annual certification process to maintain consistency in sampling techniques across lakes. Quality control is conducted after data is input into the database at the end of the sampling season and suspect measurements are flagged or removed. Part of this process identified many lakes that had unrealistic measurements prior to 2002, and were removed from this study. It is important to the Mat-Su Borough to maintain high standards for sampling and this study validated the quality of the data.

4.2 "Matanuska" and "Knik" winds as an alternative hypothesis for increasing specific conductivity in lakes

This study identified a correlation between increased specific conductivity and increased development near lakes. An alternative, but not mutually exclusive, hypothesis is that increased wind disturbance and deposition of silt plays an important role for increasing specific conductivity in lakes. There are four reasons for this hypothesis: (1) the increasing trend in wind since 1998, (2), the timing of Matanuska and Knik winds coinciding with lower than average precipitation in certain locations may allow for more aeolian transportation of silt from the braid plains of the two river valleys, (3) the location of the lakes with the highest increases in specific conductivity, and (4) episodic deposition of loess since glaciers retreated from the Mat-Su Valley.

Since 1998, when lake monitoring began, total monthly wind measured at the Matanuska Agricultural Experiment Station in Palmer for the months of May to September increased significantly overall and in each month (Figure 15).

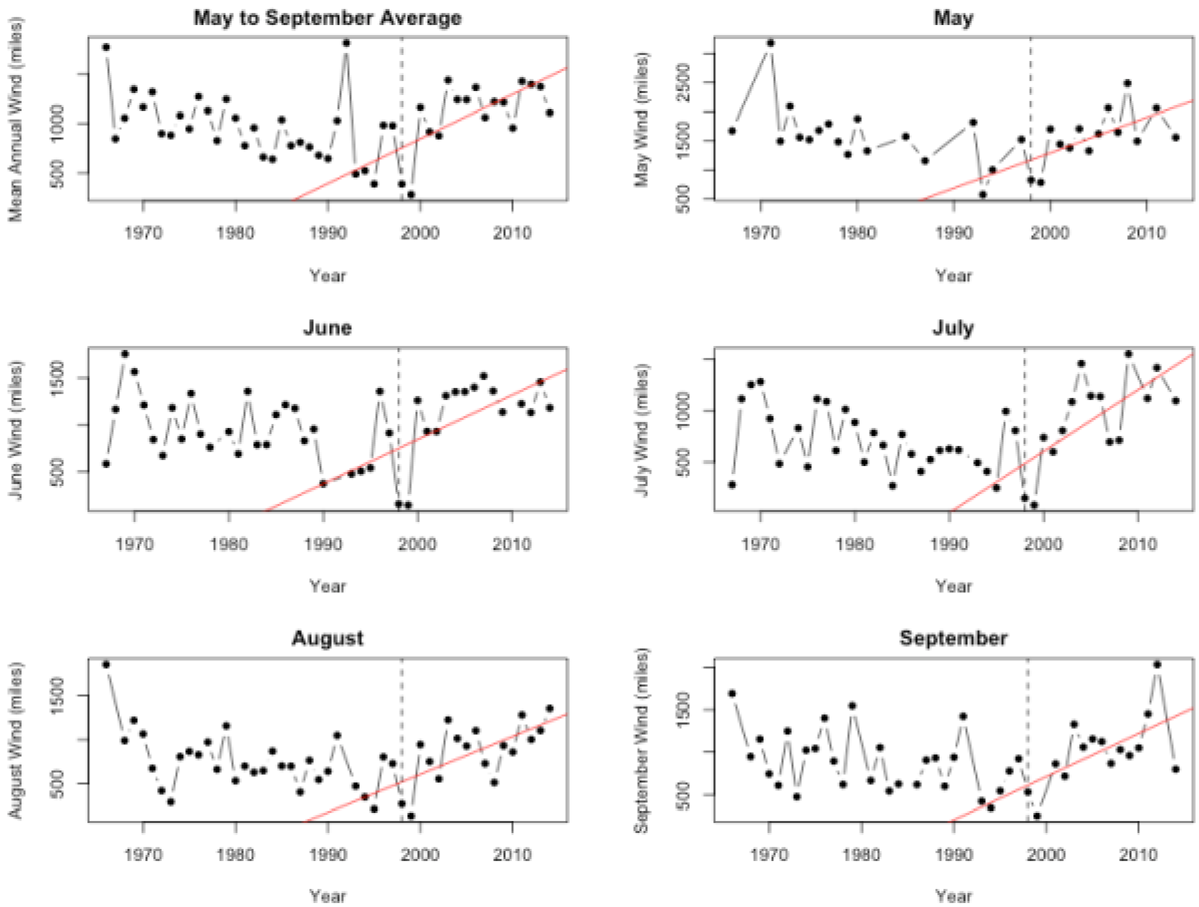

Figure 15. Total monthly wind measured at the Matanuska Agricultural Experiment Station for the months of May through September from 1966-2014. The average annual total monthly wind (upper left) from May to September shown with the linear regression trend (red) from 1998 (dashed black line) when monitoring of lake water quality began, to 2014. May through September measurements are shown in subsequent plots with trends (red) since 1998 (dashed black line) and asterisks denote significance at the p = 0.01 level.

The majority of the aeolian transportation of silt in the Matanuska-Susitna Valley comes from the Matanuska Valley (Muhs 2016). While substantial wind comes from the Knik Valley, the timing and intensity of Matanuska and Knik winds depends on the location of low pressure in the Gulf of Alaska or the Aleutian islands and high pressure in the interior of Alaska (Dale 1956). It is well known that wind is an important agent in physical limnology for creating waves, mixing, disturbance of benthos, and allochthonous inputs (Dodds 2010). In addition, silt and other fine particles can dissolve readily in lake water and contribute to measured conductance of water. Specific conductivity increased in lakes directly in the path of "Matanuska" and "Knik" winds.

Based on a geochemical analysis of Matanuska and Knik river valley silts and loess deposited in the Matanuska-Susitna Valley (such as Knik Silt Loam), Muhs et al (2016) suggested the majority of loess originates from the Matanuska Valley. Loess thins in a general westward direction and stronger winds from the Matanuska Valley coincide with low discharge in the autumn when silt-rich point bars of Matanuska River are exposed. Stronger winds in the autumn may explain higher increases in specific conductivity in August and September in all but two lakes compared to other months. The reduction in precipitation and warmer temperatures observed in Sutton, in the Matanuska Valley, may have contributed to less snow cover throughout the winter and a higher incidence of aeolian silt transportation and subsequent deposition into lakes in the Mat-Su Valley.

Considering the expected path of wind coming from the Matanuska Valley, lakes that increased in specific conductivity were located directly in the areas receiving more loess deposits over the past ~7,400 cal yr B.P (Muhs 2004). The gradual thinning of loess further west (Muhs 2004), may explain why lakes closer to the Matanuska River such as Hulbert, Jukes, Finger, Lucille, and Neklason increased in SC while others further west did not. Lakes not directly in line with the path of Matanuska or Knik winds, such as Little Question, Wolverine, and Fish, decreased in SC. Lakes further west but still in the path of Matanuska winds, Nancy, Florence, Birch and Carpenter did not change significantly in SC. The two lakes that suggest that a combination of wind and land cover change affects SC change are Seymour and Big Lake. Seymour had the most new development from 2001-2011, increased more in May than other months and, together with Big Lake, was relatively far from the wind source compared with other lakes that increased in SC.

The small decreases in pH of Seymour, Memory, Lucille, and Big Lakes, although marginally significant, support both the hypothesis that land cover change and winds are contributing factors to SC increases. Land cover change, particularly deforestation, can reduce the amount of infiltration occurring before water reaches a water body. Rain is typically acidic and if infiltration decreased, then fewer alkalizing effects from foliar exudation, leaching, and soil cation exchange occur. In a region with alkaline groundwater and surface water, sufficient infiltration for soil cation exchange is key, because soil pH is greater at depth. It is possible that the reduction in infiltration rates reduced the opportunity for precipitation (typically acidic) to interact with more alkalizing soils. Tree canopies also produce a net alkalizing affect due to foliar exudation and leaching. Deciduous trees have more of an alkalizing affect than coniferous trees on precipitation and dry deposition, and since most forest loss in the areas around lakes were deciduous forests, it is possible forest loss reduces interception of acidic precipitation and dry deposition when compared to previous to forest loss. It is also possible that increasing wind has led to more mixing, distributing more acidic water and dissolved ions throughout the water column. Careful review of the timing of stratification in the spring and fall could reveal the timing of mixing events, but the sampling most often occurred during better whether to avoid windy conditions on the lakes.

Episodic deposition of loess observed across the Mat-Su Valley can vary over very short distances (Muhs 2004). Soil formation and loess deposition are competing processes in the Matanuska-Valley and certain climate conditions may favor one over the other. It is possible that the period observed since the 1990's, while wind has been increasing and precipitation has been decreasing in some areas, is favoring loess deposition over soil formation, which occurs more rapidly in wetter conditions. This will be important for lake sediment core interpretation and understanding climate conditions that contribute to variable sedimentation rates in lakes.

Since specific conductivity (SC) is often the first measurable change in lake chemistry, increases in SC in Finger, Seymour, Memory, Lucille and Big Lake are highly suggestive of anthropogenic effects on water quality due to new development. An SC increase in Hulbert and Juke's lake without nearby development but within an increasing wind regime suggests that wind may play an important role that can increase SC due to the timing and location of winds. Further geochemical analysis and measurements of wind may help elucidate the interaction and relative roles of wind and development.

4.3 Landscape changes consisted of forest loss to residential developed land

Nearly all land change from 2001 to 2011 consisted of the removal of forest cover for residential development, and much of that was along lake shorelines. While forest loss was the most substantial change, emergent herbaceous wetlands converted to woody wetlands near Big Lake, a trend consistent with observations on the Kenai (Klein et al 2005). The conversion of forest to residential land resulted in a net increase in impervious surfaces, thus less infiltration before precipitation reaches streams and lakes, and more time for runoff to transport pollutants to surface water. It is well documented that runoff characteristics are drastically altered in watersheds with changes to land cover (Sajikumar 2014), which affects the timing and quantity of discharge into groundwater and surface water (Mateus 2014). The alteration of land cover types in favor of more developed land will continue to impact lake water quality, although the Mat-Su Borough is likely to remain rural and sparsely developed for some time. According to Brabec et al (2000), a threshold for negative impacts to water quality in lakes and streams occurs when a watershed reaches approximately 10% impervious surfaces. Most watersheds of the Mat-Su Valley have not reached that threshold but some, including Wasilla and Lake Lucille have crossed the threshold and show the expected response.

In using NLCD to document land cover changes for this study there were both advantages and disadvantages. The resolution is relatively coarse so the accuracy of classification is most likely not as high as a classification of available 1 ft. resolution imagery. Alternatively, the changes observed had to be substantial to be mapped and classified as new cover in the 2011 classification. This gives assurance that new development was consequential. Additionally, the USGS is committed to repeat classifications at least every 10 years (as Landsat imagery allows) in order to document landscape changes. Thus this study and methodology is repeatable in the Mat-Su Valley and elsewhere as long as lakes continue to be monitored. It would benefit the accuracy if a classification of land cover was conducted using higher resolution imagery, validated, and compared to the NLCD classifications for the year 2011. New imagery and data are always becoming available and will likely allow for better analysis in the future.

Other notable changes in land cover near Big Lake were the conversion of emergent wetlands to woody wetlands. The significant drying of wetlands in the Mat-Su Valley is not surprising considering findings of an increase in woody vegetation and decreasing lake areal extent in the Kenai Peninsula due to a warming climate (Klein et al. 2005). The conversion of emergent vegetation to woody vegetation reflects water as is less available to wetland plants, which indicates possible implications for water quality and quantity in lakes due to the connectivity of lakes and wetlands.

Lake front property is the highest value property in the Mat-Su Valley and it will likely remain of high value for aesthetics and recreational opportunities. There will be a limitation to how much new development can occur on lakefronts and many of the study lakes will reach a development capacity along their shorelines. Many lakes have a development plan or covenant to guide development on lakes and maintain the rural aesthetic of the Mat-Su Valley. It is possible that as development slows near lakes, there will be fewer changes to lake water quality that can be attributed to new development. It will be important to document changes and the opportunity to model future land cover and land use change can help planners avoid negative impacts to lakes.

5 Conclusion

(1) Water quality is generally good in lakes according to drinking water and aquatic habitat standards set by the EPA and ADF&G. Lakes are alkaline, have low productivity, low nutrients, only some temperatures in the summer are above standards set for salmon health, dissolved oxygen is generally good for fish, and dissolved solids remain low. The only concern that warrants investigation is within 5 meters of the bottom of Nancy Lake where specific conductivity approached 5000 $\mu S\ cm^{-1}$ for most sampling events.

(2) Geographic differences in dissolved ion chemistry reflect the dominant hydrologic input and relative distance to the Matanuska or Susitna Rivers. Specific conductivity is higher closer to the Matanuska where loess deposits have higher clay content, wind speeds are greater, and development is concentrated.

(3) Lakes located in the areas of highest development and receiving the most wind, especially shallow lakes, must continue to be carefully monitored in the future. Those lakes are Finger, Neklason, Hulbert, Juke's, Canoe, Cottonwood, Wasilla, Lucille, and to a lessor degree, Memory, and Seymour. There are many more lakes that are relatively close to these that are not monitored and consideration should be given to expanding monitoring to include those.

(4) Temporal trends in water quality were minimal in lakes and were limited to increases in specific conductivity in some lakes. An overall decrease in total

phosphorus in most lakes could be the result of more dissolved ions entering the lake and bonding with the available phosphorus, making it less available biologically and detected less. This could have cascading effects on the productivity of lakes since phosphorus is a limiting nutrient for algae.

(5) Loss in forest cover to developed land correlates with increased specific conductivity in 5 lakes, but these lakes are also located in the zone that receives higher winds and more aeolian silt transportation from Matanuska Valley winds. The highest seasonal increases in specific conductivity coincide with the heaviest rainfall and the timing of greatest transport of silt from the Matanuska braid plain. The dual anthropogenic and climatic factors may compound the increased specific conductivity but it is likely that the effect of wind is playing a larger role in increased specific conductivity, especially in shallow lakes where disturbance of the benthos is more likely with high winds.

(6) Planners who have to consider future developments and climatic changes must pay close attention to lakes that are located at the intersection of the highest wind and most development to protect aquatic habitat, recreation opportunity, and the high value aesthetics that draw people to live in the Matanuska-Susitna Valley. Outreach and education as to the threats to the lakes and the opportunities to be stewards for the lakes should be expanded by continuing the citizen-scientist led monitoring, expanding the number of metrics measured to include wind, ice out, and fecal coliform testing.

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7 Appendix

Table 8. Summary statistics of six water quality parameters measured by citizen-scientists from 1998-2000 in 46 lakes in the Matanuska-Susitna Borough.

Lake	Temperature (°C)						Dissolved Oxygen (mg L^{-1})						Specific Conductivity (µS cm^{-1})					
	Mean	Median	SD	Min	Max	n	Mean	Median	SD	Min	Max	n	Mean	Median	SD	Min	Max	n
Big Lake[a]	12.02	11.42	5.23	3.71	21.89	536	9.75	10.16	2.43	0.27	15.1	536	142.79	142.5	22.96	1	258	536
Birch[ab]	16.19	17.25	4.99	7.5	23.59	95	9.96	9.61	1.08	8.29	12.73	95	17.72	18	0.78	16	19	95
Black	16.63	16.29	2.46	13.09	19.7	18	9.3	9.65	1.94	3.83	12.3	18	257.67	253.5	15.22	242	300	18
Bruce	15.93	16.89	2.45	10.17	18.55	30	9.35	9.15	0.76	8.62	11.16	30	14.80	15	1.49	12	17	30
Canoe	9.74	10.68	4.46	4.01	18.88	34	7.1	9.22	4.84	0.13	14.3	34	487.62	481	52.40	412	566	34
Carpenter[ab]	14.69	15.94	3.51	7.54	22.3	200	9.86	9.87	1.33	1.33	13.05	200	31.98	32	2.93	25	52	200
Cornelius	10.5	10.28	3.57	5.06	16.41	32	10.19	11.24	5.33	0.79	17.13	32	238.28	218.5	38.55	202	350	32
Cottonwood	12.68	12.5	4.48	5.88	21.89	25	9.2	9.42	4.1	0.33	17.72	25	230.80	188	69.03	171	378	25
Crystal	16.49	17.74	3.45	7.28	19.9	38	8.81	9.31	2.11	1.43	11.16	38	11.55	11	5.02	9	41	38
Dandy[a]	11.16	11.24	4.58	4.26	22.1	312	8.93	9.9	3.6	0.13	13.71	312	18.64	16	15.04	12	150	312
Finger[ab]	11.06	9.93	4.52	1.71	21.55	981	7.9	9.5	4.02	0.05	15.87	981	283.69	277	48.19	129	677	981
Fish[ab]	8.5	6.6	5.05	2.4	19.61	372	4.78	3.14	4.53	0.04	12.88	372	421.10	411.5	106.64	211	718	372
Five	14.04	14.44	3.87	3.81	22.19	38	11.41	11.47	1.05	8.44	13.13	38	55.24	56	3.23	47	60	38
Florence[ab]	13.82	13.78	4.06	5.13	21.35	371	9.75	9.58	1.39	1.65	13.26	371	16.78	17	1.64	13	37	371
Hulbert[ab]	13.61	15.07	5.38	0.41	24.54	78	8.57	8.9	3.11	0.33	13.8	78	108.47	87	43.61	61	283	78
Jim	14.92	15.91	4.25	2.9	20.44	76	10.11	10.21	2	3.09	14.33	76	186.26	181	22.51	141	252	76
John	14.8	16.17	5	3.45	23.11	87	9.2	9.25	1.73	0.07	11.25	87	8.10	8	1.00	7	15	87
Juke's[ab]	11.12	11.2	2.35	6.64	15.36	42	15.16	14.49	1.7	12.48	19.17	42	319.24	321.5	15.81	282	346	42
Kalmbach	14.82	15.2	4.18	6.17	21.89	153	9.08	9.39	2.5	0.2	14.47	153	99.26	98	17.77	87	293	153
Lalen	15.23	17.96	5.84	2.8	21.64	56	10.01	9.89	1.91	3.79	14.52	56	87.95	87	14.66	76	189	56
Little Lonely	11.15	10.63	4.93	4.24	21.98	251	7.95	9.29	3.33	0.11	11.1	251	44.79	38	37.53	1	532	251
Little Question[ab]	8.29	5.57	5.31	3.43	24.09	383	4.85	1.11	4.93	0.05	14.68	383	165.21	117	199.27	2	1323	383

Long	16.35	16.97	2.35	9.2	19.39	72	8.68	8.95	1.78	0.36	13.95	72	161.49	162	3.31	156	174	72
Louise	9.41	8.31	3.62	4.29	22.16	611	9.54	9.78	1.46	0.74	12.63	611	150.98	151	6.61	131	224	611
Lucille[ab]	15.63	14.8	3.86	7.17	21.47	101	9.78	10.82	3.53	0.1	14.41	101	274.40	271	62.83	186	536	101
Lynn's	8.91	8.13	3.37	4.43	18.2	68	12.1	12.19	1.58	8.09	16.96	68	147.77	148	7.35	134	163	68
Memory[ab]	13.42	14.48	5.13	2.64	23.61	945	9.51	9.52	1.97	0.06	15.15	945	74.65	75	6.57	53	174	945
Morvro	17.2	17.74	3.23	8.42	23.02	98	9.33	9.08	1.31	6.18	19.91	98	8.88	9	0.88	6	10	98
Nancy[ab]	8.68	6.36	4.76	3.43	19.82	323	5.97	7.44	3.88	0.06	12.11	323	669.50	136	1386.86	79	5440	323
Neklason[ab]	10.11	9.45	4.65	3.4	22.19	477	9.06	10.64	5.17	0.06	22.13	477	231.35	226	34.15	181	328	477
Question	9.88	9.66	5.02	3.82	22.46	74	7.32	8.87	4.46	0.13	14.37	74	163.12	66	375.66	57.8	2060	74
Rainbow	14.78	16.67	4.99	4.24	23.03	88	8.68	8.78	1.55	5.02	11.76	88	153.09	156	11.55	131	178	88
Rocky	19.65	19.79	0.87	18.33	20.64	16	9.14	9.24	0.36	7.92	9.57	16	84.44	84.5	0.63	83	85	16
Scotty	16.4	17.27	3.78	6	23.98	37	8.96	8.68	1.32	6.96	12.73	37	15.52	16.5	3.49	1	20	37
Seventeen Mile	10.75	9.07	5.29	4.42	20.72	97	7.3	9.36	4.68	0.09	14.14	97	69.84	62	29.85	56	286	97
Seymour[ab]	16.08	16.95	4.05	6.35	23.54	394	9.12	9.4	1.84	0.35	12.68	394	150.30	150	23.12	0	292	394
Shirley	13.47	14.12	3.68	6.47	18.55	23	7.66	8.79	2.55	0.96	10.62	23	149.87	136	27.30	128	217	23
Slipper	14.26	13.56	4.32	5.74	20	25	7.43	8.18	1.94	0.72	9.06	25	798.67	769	76.00	729	928	25
Susitna	10.42	9.96	3.33	5.14	19.41	417	9.54	9.42	0.78	6.84	12.03	417	281.84	285	34.33	0	327	417
Trail	17.23	18.19	3.1	13.37	21.52	16	8.73	8.64	1.42	4.51	10.42	16	13.31	13	0.87	12	16	16
Tyone	11.34	11.81	6.25	1.8	20.6	38	10.4	10.47	1.38	7.87	13.27	38	311.05	322	56.35	0	362	38
Wasilla	11.49	11.29	4.85	0.05	19.76	315	8.19	10.07	4.19	0.07	16.23	315	255.03	243	40.63	193	468	315
West	15.17	15.91	2.92	7.95	19.36	64	9.41	9.41	1.3	4.05	12.38	64	47.36	45	8.10	42	107	64
Wolf	18.27	17.9	1.47	16.13	21.36	38	9.86	9.04	2.5	8.2	19.6	38	153.08	155	6.99	136	171	38
Wolverine[ab]	14.94	15.48	3.96	6.5	22.29	114	9.3	9.6	2.79	0.22	14.06	114	158.65	150	46.37	119	410	114
Woody	17.36	18	4.1	6.74	24.13	66	9.07	9.47	2.06	1.02	11.49	66	79.00	75	26.81	62	213	66
Mean	13.45	13.58	4.06	5.82	21.18	189.67	9.03	9.34	2.51	2.74	13.88	189.67	170.68	151.75	65.48	105.10	435.76	189.67
Median	13.93	14.46	4.14	5.10	21.54	83	9.17	9.42	1.99	0.77	13.49	83	150.08	136.00	20.14	69.00	255.00	83
SD	2.95	3.71	1.12	3.70	2.01	230.35	1.67	1.91	1.36	3.42	2.74	230.35	166.68	145.87	208.58	130.08	838.36	230.35
Minimum	8.29	5.57	0.87	0.05	15.36	16	4.78	1.11	0.36	0.04	9.06	16	8.10	8.00	0.63	0.00	10.00	16
Maximum	19.65	19.79	6.25	18.33	24.54	981	15.16	14.49	5.33	12.48	22.13	981	798.67	769.00	1386.86	729.00	5440.00	981
	pH						Chlorophyll-a (µg L^{-1})						Total Phosphorus (µg L^{-1})					

	Mean	Median	SD	Min	Max	n	Mean	Median	SD	Min	Max	n	Mean	Median	SD	Min	Max	n
Big Lake[a]	7.78	7.79	0.52	6.17	9.39	536	1.13	0.81	0.99	0.15	3.90	34	3.00	0.00	5.83	0	20	34
Birch[ab]	7.47	7.47	0.41	6.6	8.71	95	2.59	2.10	1.79	0.90	8.40	19	8.63	9.00	3.58	0.02	14	19
Black	7.73	7.96	0.50	6.31	8.25	18	-	-	-	-	-	-	-	-	-	-	-	-
Bruce	7.40	7.47	0.51	6.42	8.37	30	-	-	-	-	-	-	-	-	-	-	-	-
Canoe	7.98	8.13	0.39	7.24	8.46	34	-	-	-	-	-	-	-	-	-	-	-	-
Carpenter[ab]	7.89	7.82	0.65	6.3	9.35	200	1.59	1.60	0.79	0.60	3.30	30	7.73	7.00	2.66	4	14	30
Cornelius	8.12	8.30	0.65	6.92	8.9	32	-	-	-	-	-	-	-	-	-	-	-	-
Cottonwood	8.28	8.41	0.36	7.3	8.64	25	-	-	-	-	-	-	-	-	-	-	-	-
Crystal	7.03	7.03	0.88	5.27	8.46	38	-	-	-	-	-	-	-	-	-	-	-	-
Dandy[a]	7.38	7.51	0.59	5.88	8.79	312	1.73	1.60	0.97	0.30	3.80	34	21.82	11.50	51.62	2	283	34
Finger[ab]	7.99	8.02	0.44	6.57	9.8	981	6.42	4.20	5.26	1.10	22.00	78	21.82	20.00	11.97	0	62.5	78
Fish[ab]	7.86	7.85	0.37	6.79	8.64	372	1.56	1.20	1.23	0.00	4.70	38	12.49	11.00	7.49	2	36	38
Five	7.59	7.63	0.35	6.99	8.31	38	-	-	-	-	-	-	-	-	-	-	-	-
Florence[ab]	7.37	7.37	0.38	6.15	8.77	371	0.65	0.80	0.49	0.02	1.60	30	7.78	5.00	7.97	2	34	30
Hulbert[ab]	7.66	7.61	0.51	6.12	9.15	78	1.80	1.30	1.36	0.30	6.80	34	19.45	16.00	9.42	9	48	34
Jim	8.58	8.45	0.57	7.03	9.98	76	-	-	-	-	-	-	-	-	-	-	-	-
John	6.94	6.93	0.53	5.52	8	87	-	-	-	-	-	-	-	-	-	-	-	-
Juke's[ab]	8.28	8.31	0.36	7.09	8.95	42	0.66	0.40	0.46	0.30	1.60	13	6.80	7.00	4.21	2	14	13
Kalmbach	8.08	8.12	0.48	6.82	9.64	153	-	-	-	-	-	-	-	-	-	-	-	-
Lalen	8.36	8.58	0.62	7.02	9.19	56	-	-	-	-	-	-	-	-	-	-	-	-
Little Lonely	7.41	7.45	0.44	6.39	8.24	251	-	-	-	-	-	-	-	-	-	-	-	-
Little Question[ab]	7.28	7.21	0.50	6.24	8.59	383	2.16	2.00	1.08	1.10	5.00	26	11.53	11.00	5.67	0	24	26
Long	7.95	7.96	0.30	6.9	8.53	72	-	-	-	-	-	-	-	-	-	-	-	-
Louise	7.83	7.86	0.38	6.7	8.6	611	-	-	-	-	-	-	-	-	-	-	-	-
Lucille[ab]	8.36	8.44	0.56	6.68	9.27	101	3.89	3.95	2.44	0.00	44.90	17	21.28	21.50	11.39	0.6	9.1	17
Lynn's	8.00	7.85	0.48	7.35	9.32	68	-	-	-	-	-	-	-	-	-	-	-	-
Memory[ab]	7.73	7.67	0.58	6.27	9.93	945	2.36	1.80	1.65	0.30	9.50	121	15.08	11.50	15.61	0	158	121
Morvro	7.67	7.67	0.51	6.75	9.2	98	1.24	1.30	0.62	0.00	2.40	18	9.72	10.00	6.56	0	24	18
Nancy[ab]	7.50	7.49	0.37	6.1	8.53	323	4.21	4.00	1.92	1.50	7.90	21	11.68	13.00	7.04	2	28	21

Neklason[ab]	8.15	8.26	0.57	6.83	9.29	477	3.71	2.90	3.09	0.00	10.50	35	19.97	13.00	33.65	2	205	35
Question	7.67	7.82	0.63	6.19	8.94	74	-	-	-	-	-	-	-	-	-	-	-	-
Rainbow	7.95	7.90	0.28	7.15	8.83	88	-	-	-	-	-	-	-	-	-	-	-	-
Rocky	8.30	8.37	0.16	8.05	8.58	16	-	-	-	-	-	-	-	-	-	-	-	-
Scotty	7.10	7.13	0.48	5.95	8.06	37	-	-	-	-	-	-	-	-	-	-	-	-
Seventeen Mile	7.68	7.70	0.89	5.83	9.26	97	-	-	-	-	-	-	-	-	-	-	-	-
Seymour[ab]	8.09	8.13	0.49	5.52	10.2	394	1.67	1.30	1.03	0.50	5.10	59	10.67	10.00	7.60	0	31	59
Shirley	7.47	7.53	0.29	6.78	8.02	23	-	-	-	-	-	-	-	-	-	-	-	-
Slipper	8.22	8.26	0.33	7.49	8.79	25	-	-	-	-	-	-	-	-	-	-	-	-
Susitna	7.85	7.94	0.33	6.86	8.52	417	-	-	-	-	-	-	-	-	-	-	-	-
Trail	7.57	7.51	0.31	7.07	8.11	16	-	-	-	-	-	-	-	-	-	-	-	-
Tyone	8.18	8.21	0.28	7.6	8.65	38	-	-	-	-	-	-	-	-	-	-	-	-
Wasilla	8.12	8.21	0.49	6.95	9.06	315	-	-	-	-	-	-	-	-	-	-	-	-
West	7.33	7.31	0.33	6.64	8.02	64	-	-	-	-	-	-	-	-	-	-	-	-
Wolf	8.58	8.65	0.27	7.93	9.04	38	-	-	-	-	-	-	-	-	-	-	-	-
Wolverine[ab]	8.12	8.05	0.71	6.53	9.75	114	4.61	2.50	4.73	0.80	16.90	17	21.20	23.00	9.70	2	39	17
Woody	7.50	7.45	0.51	6.5	8.53	66	-	-	-	-	-	-	-	-	-	-	-	-
Mean	7.81	7.84	0.47	6.65	8.86	189.67	2.47	1.99	1.76	0.46	9.31	36.71	13.57	11.74	11.88	1.62	61.39	36.71
Median	7.84	7.85	0.48	6.69	8.78	83	1.80	1.60	1.23	0.30	5.10	30	11.68	11.00	7.60	2.00	31.00	30
SD	0.40	0.42	0.15	0.60	0.56	230.35	1.53	1.19	1.20	0.36	7.81	24.14	7.58	6.72	9.41	1.55	55.29	24.14
Minimum	6.94	6.93	0.16	5.27	8.00	16	0.65	0.40	0.46	0.00	1.60	13	3.00	0.00	2.66	0.00	9.10	13
Maximum	8.58	8.65	0.89	8.05	10.20	981	6.42	4.20	5.26	1.50	44.90	121	21.82	23.00	51.62	9.00	283.00	121

[a] -- Lakes with ≥ 5 years of data with ≥ 3 months of sampling during the ice free season in each year used in the temporal trend analysis.

[b] – Lakes with geographically independent watersheds meeting temporal trend analysis criteria used for geospatial trend analysis.